HOME LAB

Small-business infrastructure — complete rebuild guide (virtual or hardware)

Active Directory · GPO · AGDLP file server · OPNsense *or* Cisco router + switch, with DMZ · GLPI help desk · NAS with RAID 1 · backups with restore test · Wazuh SIEM · internal PKI (AD CS) with LDAPS and trusted certificates

Item	Value
Domain	lab.local (NetBIOS: LAB)
Address block	10.2.0.0/24 split with VLSM into 4 zones
Network, track A	VMware Workstation on a Windows 11 laptop + OPNsense (as built)
Network, track B	Cisco router + managed switch + Proxmox server (same IP plan, CCNA practice)
Virtual machines	9 (firewall, domain controller, certificate authority, SIEM, NAS, file server, GLPI server, reverse proxy, client)
Based on	the TSSR “NovaTech” project, rebuilt alone at home
Version	1.1 — October 2026 (v1.1: PKI added)

Note — No password is written in this document. Every account points to its entry in the password manager (KeePass/Bitwarden). MAC addresses are those of the current build: VMware generates new ones if you rebuild the VMs, so update the inventory table after a rebuild.

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How to use this guide

This guide lets you rebuild the whole lab **from an empty computer**, in the right order, and understand why each choice was made.

Pick your network track

	Track A — full virtualisation	Track B — real hardware
Read	Part 2 (chapters 1A–2A)	Part 3 (chapters 1B–12B)
Firewall / routing	OPNsense VM (stateful firewall)	Cisco router: sub-interfaces, ACLs, NAT
VLANs	VMware networks (VMnet1, LAN segments)	802.1Q VLANs on a managed Cisco switch
Servers run on	VMware Workstation (laptop)	Proxmox VE (VLAN-aware bridge on a trunk)
Cost	free, one PC with 32 GB RAM	≈ €150–300 used gear + a PC for Proxmox
Best for	learning the whole lab quickly	CCNA practice, a real cabled network

Both tracks give the **same IP plan, gateways, zones and rules**. After your network track, continue with **Part 4** (servers), which is shared: the opening table of Part 4 translates the few VMware words for Proxmox. Mixed setups also work — for example OPNsense on a small PC with the Cisco switch (Appendix C).

How each chapter is written

Element	What it gives you
Intro (2–3 lines)	what you are building and why
Numbered steps	exactly where to click and what to type; Windows/OMV labels are shown **in French in bold**, as they appear on screen
Settings tables	field → value, ready to copy
Code blocks	commands to paste as-is (replace values in <angle brackets>)
Check boxes	the test that proves the step works, with the expected result
Watch out boxes	real traps met while building this lab
Why boxes	the concept behind a choice (useful for exams and interviews)

Build order

#	Chapter	Depends on
1A–2A	Track A: host PC, VMware networks, OPNsense	—
1B–12B	Track B: cabling, switch, router, NAT, ACLs, Proxmox, tests	—
3	SRV-AD: domain controller, DNS, DHCP, NTP	your track
4	AD structure: OUs, users, groups (AGDLP)	3
5	FS01: file server and shares	4
6	GPOs and logon hours	5
7	CLT01: client workstation and tests	6
8	LNX01: GLPI, LDAP, SSH by key, ufw	3
9	NAS01: OpenMediaVault, RAID 1	3
10	Backups and restore test	5, 8, 9
11	SIEM01: Wazuh, agents, firewall/router syslog	track, 3
12	DMZ: PROXY01, Nginx HTTPS, port forwarding	8, 11
13	GLPI in use: profiles, tickets, knowledge base	8
14	PKI: CA01, templates, auto-enrollment, LDAPS, trusted certificates, CA monitoring	3, 8, 11, 12
15–19	Tests, procedures, known problems, policies, improvements	all

Part 0 — Overview

Scope, hardware, and the three diagrams of the lab.

Scope and hardware

What the lab provides

Service	Machine	Technology
Routing, filtering, NAT, DHCP relay, syslog	OPNsense (fw01)	OPNsense 26.7
Directory, DNS, DHCP, time, GPO	SRV-AD	Windows Server 2022
Certificate authority (PKI): LDAPS, HTTPS certificates	CA01	Windows Server 2025 · AD CS
File shares with AGDLP permissions	FS01	Windows Server 2025
Help desk and knowledge base	LNX01	Debian 13 · GLPI 12 · Apache · MariaDB
Backup storage (RAID 1)	NAS01	OpenMediaVault (Debian 13)
Security monitoring (SIEM)	SIEM01	Ubuntu Server 26.04 · Wazuh 4.14.9
Publication of GLPI over HTTPS	PROXY01 (DMZ)	Debian 13 · Nginx
User workstation	CLT01	Windows 11 Pro

Host computer

Item	Value	Note
Model	Lenovo IdeaPad Pro 5	
CPU	AMD Ryzen 7 8845HS — 8 cores / 16 threads	AMD-V enabled; CPU is never the bottleneck (≈5% with 8 VMs)
RAM	32 GB LPDDR5X 7500 (soldered) — 30.8 GB usable	BIOS: **UMA Frame Buffer Size = 1G** (default 4G) gives back ≈4 GB
Disk	1 TB NVMe SSD	VM disks thin-provisioned
Hypervisor	VMware Workstation Pro (free for personal use)	

RAM budget

Running all 8 VMs fills the laptop (≈97% RAM at idle). Work **by scenario**: base = OPNsense + SRV-AD, then add only what the task needs.

VM	RAM	vCPU	Disk(s)
01 OPNsense	1–2 GB	2	20 GB
02 SRV-AD	3–4 GB	2	60 GB
03 SIEM01	6–8 GB	4	60 GB (all to /)
04 NAS01	1.5–2 GB	2	16–20 GB system + 2 × 50 GB data
05 FS01	3–4 GB	2	60 GB system + 40 GB data (D:)
06 LNX01	2 GB	2	20 GB
07 PROXY01	1 GB	1	10 GB
08 CLT01	3–4 GB	2	64 GB (vTPM, encrypted VM)
09 CA01	2–3 GB	2	60 GB
Total	**≈ 23–30 GB**		

Scenario	VMs to start	≈ RAM
Base (always)	OPNsense, SRV-AD	4–6 GB
Shares / GPO tests	+ FS01, CLT01	10–14 GB
GLPI	+ LNX01 (+ CLT01)	6–10 GB
Backups	+ NAS01 + servers to back up	8–14 GB
SIEM	+ SIEM01 + monitored machines	10–16 GB
DMZ	+ LNX01 + PROXY01 (+ SIEM01)	7–13 GB
PKI	+ CA01 (+ LNX01, PROXY01, SIEM01 for their certificates)	7–18 GB

Diagrams

Logical diagram

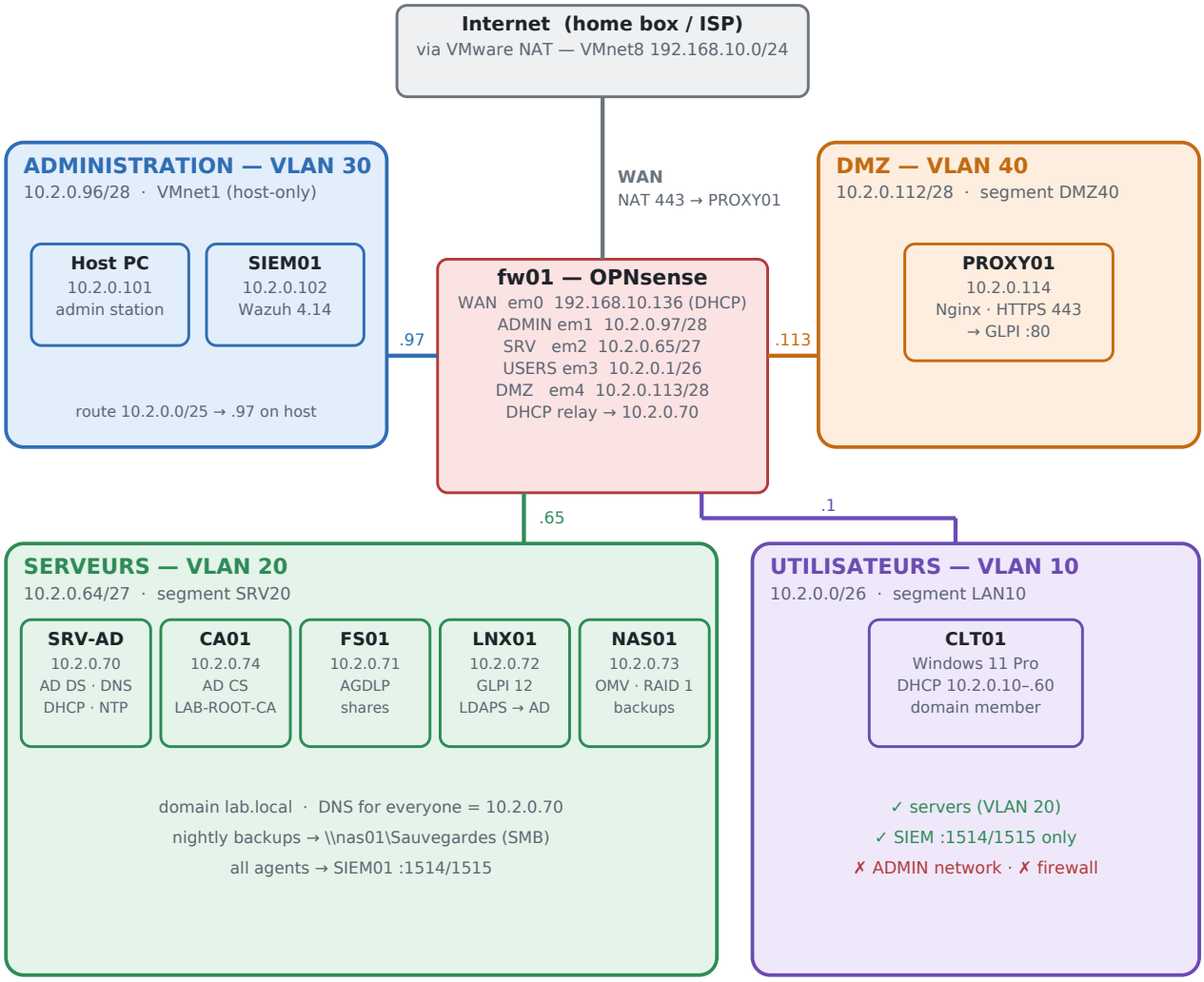


Figure 1 — Logical view: zones, addresses, and what each zone may reach.

Physical diagram — home lab (VMware)

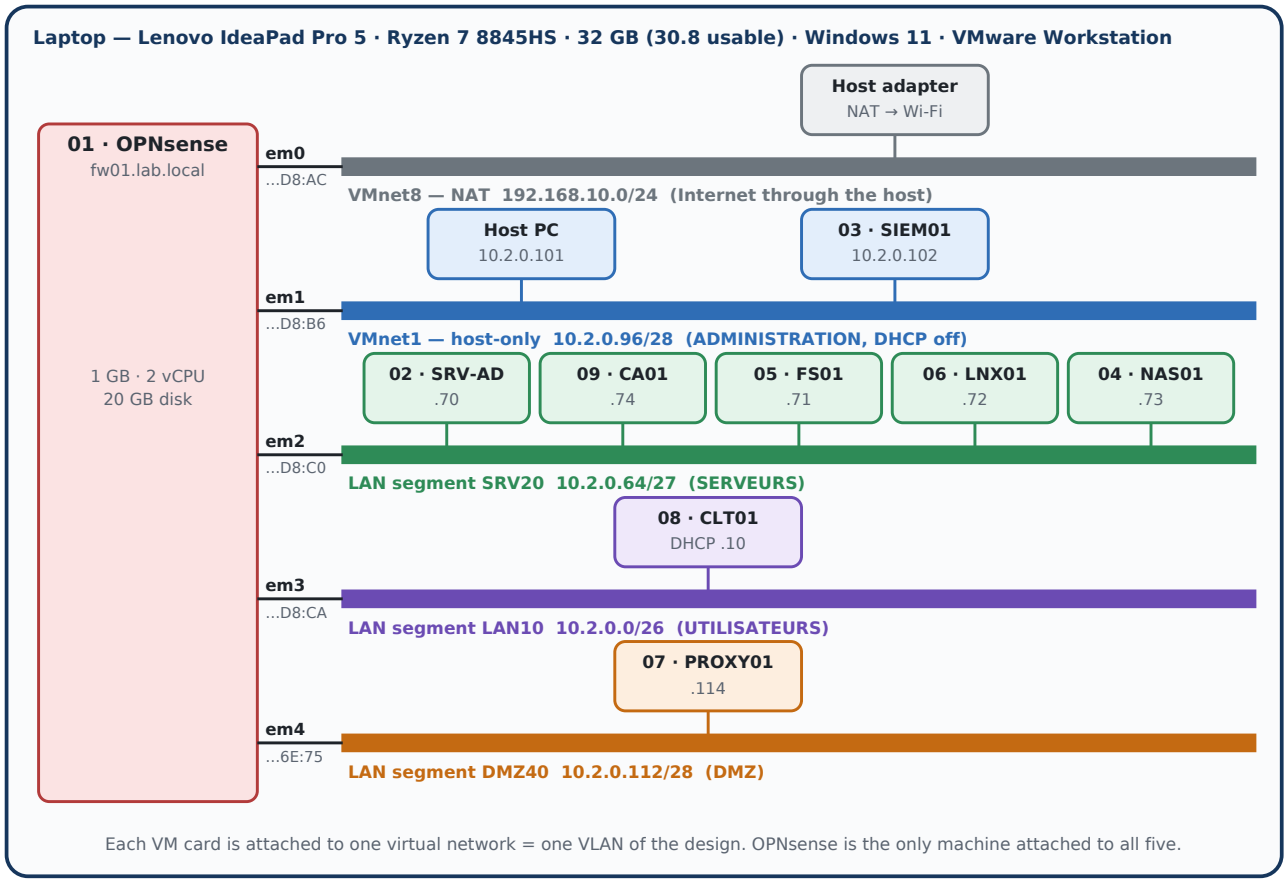
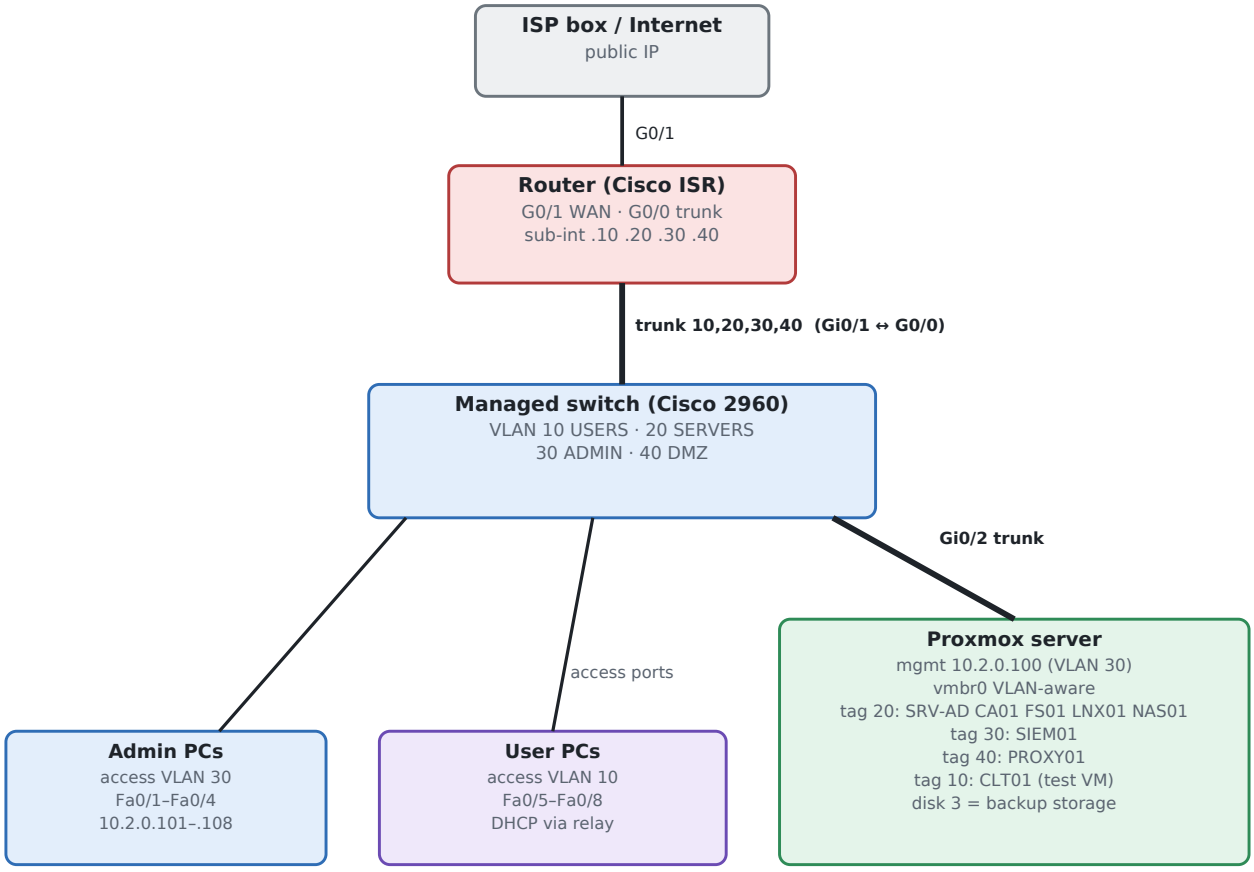


Figure 2 — Physical view of the home lab: each virtual network plays the role of one VLAN.

Physical diagram — real hardware (track B)



Same IP plan as the VMware lab — only the transport of the VLANs changes.

Figure 3 — Track B: Cisco router (router-on-a-stick), managed switch, Proxmox server.

Part 1 — Reference sheets

Everything you need to find an address, a link or an account in seconds.

Addressing plan (VLSM)

Block **10.2.0.0/24**, split largest need first so every subnet stays aligned. Gateways are the **first usable address** of each subnet (they are OPNsense's interfaces).

Zone / VLAN	Need	Network	Mask	Usable range	Gateway	Broadcast
UTILISATEURS — 10	50	10.2.0.0/26	255.255.255.192	.1 – .62	10.2.0.1	10.2.0.63
SERVEURS — 20	20	10.2.0.64/27	255.255.255.224	.65 – .94	10.2.0.65	10.2.0.95
ADMINISTRATION — 30	10	10.2.0.96/28	255.255.255.240	.97 – .110	10.2.0.97	10.2.0.111
DMZ — 40	≤ 14	10.2.0.112/28	255.255.255.240	.113 – .126	10.2.0.113	10.2.0.127
Reserve	—	10.2.0.128/25	255.255.255.128	.129 – .254	—	10.2.0.255
WAN (VMware NAT)	—	192.168.10.0/24	255.255.255.0	DHCP by VMware	192.168.10.2	—

DHCP

Scope	Range	Options	Served by
VLAN10-Utilisateurs (10.2.0.0/26)	10.2.0.10 – 10.2.0.60	003 Router 10.2.0.1 · 006 DNS 10.2.0.70 · 015 Domain lab.local	SRV-AD, through OPNsense DHCP relay

Servers, admin and DMZ machines all use **static** addresses. Addresses .2–.9 and .61–.62 of VLAN 10 are kept free for printers or other static devices.

Machine inventory

Identity and resources

#	VM (VMware)	Hostname / FQDN	OS	Role	VMware folder
01	01 - OPNSense	fw01.lab.local	OPNsense 26.7	firewall, router, NAT, relay	01-Infrastructure
02	02 - SRV-AD	SRV-AD.lab.local	Win Server 2022 Std	AD DS, DNS, DHCP, NTP	01-Infrastructure
03	03 - SIEM01	siem01.lab.local	Ubuntu Server 26.04	Wazuh SIEM	02-Serveurs
04	04 - NAS01	nas01.lab.local	OpenMediaVault	backup storage, RAID 1	02-Serveurs
05	05 - FS01	FS01.lab.local	Win Server 2025 DC	file server (AGDLP)	02-Serveurs
06	06 - LNX01	lnx01.lab.local	Debian 13	GLPI 12	02-Serveurs
07	07 - PROXY01	proxy01.lab.local	Debian 13 (no GUI)	Nginx reverse proxy	03-DMZ
08	08 - CLT01	CLT01.lab.local	Windows 11 Pro	client workstation	04-Clients
09	09 - CA01	CA01.lab.local	Win Server 2025 DC	Enterprise root CA (AD CS)	01-Infrastructure

Network

VM	Interface	IP / prefix	Gateway	DNS	MAC address	VMware network
OPNsense	em0 WAN	DHCP (192.168.10.136)	192.168.10.2	—	00:0C:29:9D:D8:AC	NAT (VMnet8)
	em1 ADMIN	10.2.0.97/28	—	—	00:0C:29:9D:D8:B6	VMnet1 host-only
	em2 SERVEURS	10.2.0.65/27	—	—	00:0C:29:9D:D8:C0	segment SRV20
	em3 UTILISATEURS	10.2.0.1/26	—	—	00:0C:29:9D:D8:CA	segment LAN10
	em4 DMZ	10.2.0.113/28	—	—	00:50:56:2F:6E:75	segment DMZ40
SRV-AD	Ethernet0	10.2.0.70/27	10.2.0.65	127.0.0.1	00:0C:29:B4:64:C2	SRV20
FS01	Ethernet0	10.2.0.71/27	10.2.0.65	10.2.0.70	00:0C:29:32:B6:2C	SRV20
LNX01	ens33	10.2.0.72/27	10.2.0.65	10.2.0.70	00:0C:29:BD:24:24	SRV20
NAS01	ens33	10.2.0.73/27	10.2.0.65	10.2.0.70	00:0C:29:61:1E:46	SRV20
SIEM01	ens33	10.2.0.102/28	10.2.0.97	10.2.0.70	00:0C:29:3A:97:99	VMnet1
PROXY01	ens33	10.2.0.114/28	10.2.0.113	10.2.0.70	00:0C:29:8D:29:D0	DMZ40
CLT01	Ethernet0	DHCP (10.2.0.10)	10.2.0.1	10.2.0.70	00:0C:29:BC:25:63	LAN10
CA01	Ethernet0	10.2.0.74/27	10.2.0.65	10.2.0.70	00:0C:29:F1:DB:C3	SRV20
Host PC	VMnet1 adapter	10.2.0.101/28	none	home box	(host)	VMnet1

Track B network devices

Device	Interface	IP / prefix	Access
R-LAB (router)	G0/0.30 · .20 · .10 · .40	10.2.0.97 · .65 · .1 · .113	SSH admin@10.2.0.97 (from ADMIN only)
	G0/1 (outside)	DHCP from the home box	—
SW-LAB (switch)	VLAN 30 SVI	10.2.0.98/28, gateway .97	SSH admin@10.2.0.98
Proxmox (pve)	vmbr0 (VLAN 30)	10.2.0.100/28, gateway .97	https://10.2.0.100:8006 (root@pam)
Admin PC	switch access port VLAN 30	10.2.0.101/28, gateway .97	—

Note — On track B the servers keep the IPs above; their MACs change (Proxmox generates BC:24:11:... addresses). MACs starting with **00:0C:29** are generated by VMware when a VM is created; **00:50:56** is another VMware range (used here for the DMZ card added later). Both are normal.

DNS records (zone lab.local and reverse 0.2.10.in-addr.arpa)

Name	Type	Address	PTR	Created by
srv-ad	A	10.2.0.70	yes	automatic (DC)
FS01, CLT01, CA01	A	10.2.0.71 · 10.2.0.10 · 10.2.0.74	yes	automatic (domain members)
lnx01	A	10.2.0.72	yes	manual
nas01	A	10.2.0.73	yes	manual
siem01	A	10.2.0.102	yes	manual
proxy01	A	10.2.0.114	yes	manual

Web interfaces and links

Watch out — The host PC uses the **home box** as DNS server, so it does not know **lab.local** names. From the host PC, always use the **IP addresses**. Domain machines (CLT01, servers) can use the names. Since chapter 14, the host PC trusts **LAB-ROOT-CA** (imported by hand, it is not a domain member): GLPI, OPNsense and Wazuh open with a padlock.

Service	URL from the host PC (admin)	URL from inside the lab	Account (login source)	Reachable from
OPNsense	https://10.2.0.97	—	root	ADMIN only (GUI listens on ADMINISTRATION only) · certificate fw01.lab.local
Wazuh dashboard	https://10.2.0.102	https://siem01.lab.local	admin	ADMIN only · certificate siem01.lab.local
OpenMediaVault	http://10.2.0.73	http://nas01.lab.local	admin (web)	ADMIN, SERVEURS
GLPI (internal)	http://10.2.0.72	http://lnx01.lab.local	glpi (Base interne GLPI) · AD users (AD lab.local)	all internal zones
GLPI (published)	https://192.168.10.136 (or https://glpi.lab.local with a hosts entry)	—	same as above	“Internet” side, through OPNsense NAT → PROXY01 → LNX01 · certificate glpi.lab.local
CA (CRL, CA certificate)	http://10.2.0.74/CertEnroll/	http://ca01.lab.local/CertEnroll/	anonymous read	internal zones (not the DMZ)
CA console	certsrv.msc on CA01	—	LAB\Administrateur	admins only
File shares	\\FS01 (from domain PCs)	\\FS01\Commun …	AD users	domain members
Backup share	\\nas01\Sauvegardes	\\10.2.0.73\Sauvegardes	nas01\SVC-Backup	servers, admin
Admin share FS01	\\FS01\D$\PARTAGES		LAB\Administrateur	admins only

SSH access

Server	Command (from host PC)	Authentication	Rules
LNX01	ssh tech1@10.2.0.72	SSH key only	root login off, password off; ufw: 22 from 10.2.0.96/28 only
SIEM01	ssh nicolas@10.2.0.102	password (+ key optional)	admin network only
PROXY01	ssh nicolas@10.2.0.114	password (+ key optional)	from ADMIN only (DMZ cannot initiate)
NAS01	ssh <user>@10.2.0.73	member of group _ssh	root login off (or SSH disabled)

Accounts

Every password is stored in the password manager, under **Lab › <machine> › <account>**.

Account	Type	Where	Used for
root	local	OPNsense	web GUI and console
LAB\Administrateur	domain admin	AD	administration of Windows servers and AD
.\Administrateur	local	each Windows machine	local admin (after the domain join, prefix .\)
DSRM	recovery	SRV-AD	Directory Services Restore Mode
13 users (prenom.nom)	domain users	OU UTILISATEURS\...	business accounts (see chapter 4)
test.recette	domain user	OU INFORMATIQUE	lockout / password policy tests
SVC-Backup	service (domain)	OU COMPTES-ADMINISTRATION	Windows Server Backup (Opérateurs de sauvegarde)
SVC-Backup	local	NAS01 (OMV)	write access to Sauvegardes — **same password** as the domain one
svc-glpi	service (domain)	OU COMPTES-ADMINISTRATION	GLPI LDAP bind (read-only, never expires)
glpi	GLPI local	LNX01 / GLPI	GLPI super-admin (Base interne GLPI)
tech, normal, post-only	GLPI local	GLPI	default accounts — **disabled**
maxime.roux, clara.garnier	AD → GLPI	GLPI	profiles Technician + Self-Service
lnxadmin / tech1	local Linux	LNX01	console / SSH by key (sudo)
root	local Linux	LNX01	MariaDB root via sudo
glpi (MariaDB)	database	LNX01	GLPI database user (localhost only)
nicolas	local Linux	SIEM01, PROXY01	admin (sudo)
admin	Wazuh	SIEM01 dashboard	generated at install (kept)
root / admin	local / web	NAS01	console / OMV web interface
(VM encryption)	VMware	CLT01	password of the encrypted VM (vTPM)
(config backup)	OPNsense	exported XML	encryption password of the backup file
(BitLocker key)	host PC	Microsoft account	recovery key of the laptop
(CA backup)	file password	CA01 · \\nas01\Sauvegardes\CA01	protects LAB-ROOT-CA.p12 (CA private key)

Ports and flows

Only these flows cross the firewall between zones. Everything else is denied (default deny).

From	To	Port / proto	Purpose	OPNsense rule
UTILISATEURS	any (relay)	UDP 67	DHCP requests	DHCP relay
UTILISATEURS	SIEM01	TCP 1514, 1515	Wazuh agent	Agents Wazuh vers SIEM01
UTILISATEURS	ADMINISTRATION	any	**blocked + logged**	Bloque Utilisateurs vers Admin
UTILISATEURS	This Firewall	any	**blocked + logged**	Bloque accès au pare-feu
UTILISATEURS	any	any	servers + Internet	Utilisateurs vers tout
SERVEURS	any	any	servers out, agents, NTP, backups	Serveurs vers tout
ADMINISTRATION	any	any	administration	default allow LAN
DMZ	SRV-AD	TCP/UDP 53	DNS	DMZ vers DNS
DMZ	LNX01	TCP 80	proxy → GLPI	DMZ vers GLPI
DMZ	SIEM01	TCP 1514, 1515	Wazuh agent	DMZ vers SIEM
DMZ	10.2.0.0/24	any	**blocked + logged**	Bloque DMZ vers le lab
DMZ	any	TCP 80, 443	updates	DMZ vers Internet (web)
WAN	PROXY01	TCP 443	GLPI publication	Destination NAT + linked rule
OPNsense	SIEM01	UDP 514	syslog	(firewall's own traffic)
Agents (all)	SIEM01	TCP 1514/1515	events / enrolment	see above
SRV-AD, FS01, LNX01, CA01	NAS01	TCP 445	backups (SMB)	same zone, no rule
LNX01 (GLPI)	SRV-AD	TCP 636	LDAPS bind (encrypted)	same zone, no rule
Domain members	CA01	TCP 135 + RPC	certificate enrollment	same zone / Utilisateurs vers tout
Internal clients	CA01	TCP 80	CRL and CA certificate download	same zone / existing rules
SRV-AD	pool.ntp.org	UDP 123	time source of the domain	Serveurs vers tout

Operating the lab

VMware library

```
Home Lab
  ├─ 01-Infrastructure : 01 - OPNSense, 02 - SRV-AD, 09 - CA01
  ├─ 02-Serveurs       : 03 - SIEM01, 04 - NAS01, 05 - FS01, 06 - LNX01
  ├─ 03-DMZ            : 07 - PROXY01
  └─ 04-Clients        : 08 - CLT01
```

VM files live in **C:\VMs**, never in a OneDrive-synced folder. Keep school VMs (Kali, Metasploitable, Proxmox) outside this library folder. **Metasploitable must stay on its own isolated segment**, never on NAT or bridged networks.

Start-up and shut-down order

Start	VM	Wait before next	Why
1	OPNsense	≈ 1 min	routing between zones, Internet
2	SRV-AD	2–3 min	DNS, AD, DHCP, time — everyone depends on it
3	CA01	≈ 1 min	certificate enrollment, CRL publication
4	SIEM01	2–3 min	ready to receive logs from the others
5	NAS01	≈ 1 min	backup target
6	FS01	≈ 1 min	needs AD
7	LNX01	≈ 1 min	GLPI needs AD/DNS for LDAPS; Apache/MariaDB start slowly
8	PROXY01	30 s	needs LNX01
9	CLT01	—	last: domain, shares, GPOs available

Shut down in reverse order (CLT01 → OPNsense), always cleanly: Windows **Démarrer → Arrêter**, Linux **sudo poweroff**, OPNsense console option **5**, OMV **power icon → Arrêter**. Avoid VMware “power off” except for a frozen VM (risk for MariaDB, the Wazuh indexer and AD).

> **Note —** Scheduled backups (21:00–22:00) only run if the VMs are on at that time.

Part 2 — Network, track A: VMware + OPNsense

The lab as built on the laptop: VMware virtual networks and an OPNsense firewall. Skip to Part 3 if you build on real hardware.

1A. Host PC preparation

The laptop plays the role of the hardware: it hosts the virtual networks (the “switch”) and is also the **admin workstation**, plugged into the ADMINISTRATION network.

1A.1 Install VMware Workstation and free some RAM

1. Install **VMware Workstation Pro** (free for personal use; download from the Broadcom portal, choose the *Personal* edition).
2. BIOS: **F2** at boot (or Windows **Paramètres → Système → Récupération → Démarrage avancé → Dépannage → Options avancées → Paramètres du microprogramme UEFI**). In **Configuration**, set **UMA Frame Buffer Size = 1G**, then **F10**.
3. Before touching the BIOS, note the **BitLocker recovery key** (account.microsoft.com/devices/recoverykey). Check which key belongs to this PC with `manage-bde -protectors -get C:`.

Check — Task Manager → **Performances → Mémoire**: usable RAM ≈ 30.8 GB instead of ≈ 27 GB.

1A.2 Virtual networks

Edit → Virtual Network Editor (Modifier → Éditeur de réseau virtuel), **Modifier les paramètres** (admin):

Network	Type	Settings	Role
VMnet8	NAT	subnet 192.168.10.0/24, DHCP on (default)	WAN of OPNsense = “Internet”
VMnet1	Host-only	subnet **10.2.0.96**, mask **255.255.255.240**, **untick** “Utiliser le service DHCP local”, keep “Connecter un adaptateur hôte”	ADMINISTRATION (VLAN 30)
SRV20	LAN segment	created from a VM card (Segments LAN… → Ajouter)	SERVEURS (VLAN 20)
LAN10	LAN segment	idem	UTILISATEURS (VLAN 10)
DMZ40	LAN segment	idem	DMZ (VLAN 40)

Why — A LAN segment is a private virtual switch with no host access: perfect to emulate an isolated VLAN. Only ADMINISTRATION is host-only, because the laptop itself must sit in the admin network.

1A.3 Host adapter IP and route

1. `ncpa.cpl` → **VMware Network Adapter VMnet1 → Propriétés → TCP/IPv4**: IP **10.2.0.101**, mask **255.255.255.240**, **no gateway**, **no DNS**.
2. cmd **as administrator** — persistent route to the whole lab through OPNsense:

```
route -p add 10.2.0.0 mask 255.255.255.128 10.2.0.97
```

Why — The adapter has no gateway on purpose (home Internet must not go through the lab). Without the route, traffic to 10.2.0.70 leaves through the home box and the ISP answers “destination network unreachable”. `-p` keeps the route after reboots; you add it only once. 10.2.0.0/25 covers the four zones (.0 to .127).

Check — `route print 10.2.0.0` shows the route under **Itinéraires persistants**. If it ever disappears: a VPN (ExpressVPN) or a reset of VMnet1 — just add it again.

2A. OPNsense firewall (fw01)

OPNsense replaces the Cisco router and the ACLs of the school project: one network card per zone, routing between them, filtering, NAT to the Internet, DHCP relay, logging to the SIEM.

2A.1 VM

Setting	Value
Guest OS	Other → FreeBSD 14 64-bit
CPU / RAM / disk	2 vCPU · 1–2 GB · 20 GB
Card 1	NAT (VMnet8) → em0 WAN
Card 2	Personnalisé: VMnet1 → em1 ADMINISTRATION
Card 3	Segment LAN SRV20 → em2 SERVEURS
Card 4	Segment LAN LAN10 → em3 UTILISATEURS
Card 5	Segment LAN DMZ40 → em4 DMZ
ISO	OPNsense dvd image (amd64), .iso.bz2 extracted with 7-Zip

Watch out — Note each card's **MAC** (card → **Avancé…**). OPNsense names cards em0…em4; the MAC is the only reliable way to know which is which. Remove the CD drive after install: an empty drive floods the logs with devd … `cd0` errors.

2A.2 Installation and console

1. Boot, log in `installer` / `opnsense`, keymap French, **Install (UFS)** on the 20 GB disk, set the **root** password, reboot, disconnect the ISO.
2. Log in as root. Option **1) Assign interfaces**: LAGG **n**, VLAN **n**, then WAN `em0`, LAN `em1`, OPT1 `em2`, OPT2 `em3`, OPT3 `em4`, Enter, **y**. (Ctrl+C cancels the wizard without saving.)
3. Option **2) Set interface IP address**, once per interface (DHCP **n**, no upstream gateway, IPv6 **n**, DHCP server **n**, keep HTTPS GUI):

Interface	Card	IPv4	Bits
LAN	em1	10.2.0.97	28
OPT1	em2	10.2.0.65	27
OPT2	em3	10.2.0.1	26
OPT3	em4	10.2.0.113	28

Check — From the host: `ping 10.2.0.97`. The console shows WAN with a 192.168.10.x address.

2A.3 Web wizard and updates

1. Browse **https://10.2.0.97** (accept the self-signed certificate), log in **root**.
2. Wizard: hostname `fw01`, domain `lab.local`, timezone **Europe/Paris**; WAN = DHCP; LAN keep 10.2.0.97/28 and **untick Configure DHCP server**; untick **Optimize for Multiwan** and **Automatic DHCP/DNS registration**; root password empty (unchanged); Apply.
3. **System → Firmware → Status → Check for updates → Update** (reboot if asked).
4. **Interfaces → [OPT1/OPT2/OPT3/LAN]**: tick Enable, Description = **SERVEURS**, **UTILISATEURS**, **DMZ**, **ADMINISTRATION** → Save → Apply changes.
5. **System → Settings → General**: Timezone **Europe/Paris**.

2A.4 Aliases

Firewall → Aliases → +. **Choose the Type first**, then type each value and press Enter so it becomes a tag. Save each alias, then **Apply**.

Name	Type	Content	Description
DHCP_SERVER	Port(s)	67	Port DHCP serveur
SRV_AD	Host(s)	10.2.0.70	Contrôleur de domaine
LNX01	Host(s)	10.2.0.72	Serveur GLPI
SIEM01	Host(s)	10.2.0.102	Serveur Wazuh
LAB_NETS	Network(s)	10.2.0.0/24	Tous les réseaux du lab
PORT_DNS	Port(s)	53	DNS
PORT_HTTP	Port(s)	80	HTTP
PORT_WEB	Port(s)	80, 443	Web sortant
WAZUH_PORTS	Port(s)	1514, 1515	Ports agents Wazuh

Watch out — If the Type stays on **Host(s)**, port numbers are rejected (“not a valid hostname, IP address or range”).

2A.5 Firewall rules

Firewall → Rules, choose the interface in the selector, red **+**. For every rule: Direction **In**, Version **IPv4**, Quick ticked. Rules are read **top to bottom, first match wins**; reorder with the arrows in **Commands**, then **Apply**. In 26.x the source “net” is called **<zone> network**.

ADMINISTRATION

Keep the automatic **Default allow LAN to any** rules (and the anti-lockout rule).

SERVEURS

#	Action	Proto	Source	Destination	Port	Description
1	Pass	any	SERVEURS network	any	any	Serveurs vers tout

UTILISATEURS

#	Action	Proto	Source	Destination	Port	Log	Description
1	Pass	UDP	any	any	DHCP_SERVER		DHCP relay
2	Pass	TCP	UTILISATEURS network	SIEM01	WAZUH_PORTS		Agents Wazuh vers SIEM01
3	Block	any	UTILISATEURS network	ADMINISTRATION network	any	✓	Bloque Utilisateurs vers Admin
4	Block	any	UTILISATEURS network	This Firewall	any	✓	Bloque accès au pare-feu
5	Pass	any	UTILISATEURS network	any	any		Utilisateurs vers tout

Why — Rule 2 is a **narrow exception** (one host, two ports) placed above the block. Rule 4 replaces the Cisco `access-class` on vty lines: without it, CLT01 could open the firewall GUI on 10.2.0.1.

DMZ

#	Action	Proto	Source	Destination	Port	Log	Description
1	Pass	TCP/UDP	DMZ network	SRV_AD	PORT_DNS		DMZ vers DNS
2	Pass	TCP	DMZ network	LNX01	PORT_HTTP		DMZ vers GLPI
3	Pass	TCP	DMZ network	SIEM01	WAZUH_PORTS		DMZ vers SIEM (agent Wazuh)
4	Block	any	DMZ network	LAB_NETS	any	✓	Bloque DMZ vers le lab
5	Pass	TCP	DMZ network	any	PORT_WEB		DMZ vers Internet (web)

Why — Rule 4 comes **before** rule 5, so “any” in rule 5 can only mean the Internet. DNS uses UDP and falls back to TCP for large answers, hence TCP/UDP.

2A.6 DHCP relay

1. **Services → DHCRelay**. Top table **Destinations → +**: Name `SRV-AD`, Server `10.2.0.70` → Save.
2. Bottom table **Relays → +**: Enabled ✓, Interface **UTILISATEURS**, Destination **SRV-AD** → Save → Apply.

> **Note —** Equivalent of `ip helper-address 10.2.0.70` on a Cisco sub-interface. No DHCP server may run on the same interface (Kea/Dnsmasq stay disabled).

2A.7 Logging, GUI exposure and WAN options

Menu	Setting	Value	Why
Firewall → Settings → Advanced	Log default block / bogons / private	✓	unexpected traffic is visible
	Log default pass	✗	removes NTP / anti-lockout noise
	Log automatic source NAT	✗	noise
System → Settings → Administration	Listen Interfaces	**ADMINISTRATION only**	the GUI must never answer on the WAN; frees port 443 for the DMZ publication
Interfaces → [WAN]	Block private networks	✗ (lab only)	the lab WAN is itself private (VMware NAT); keep ✓ in production
Interfaces → [DMZ]	Block private / bogon	✗	internal zone

2A.8 Remote syslog and configuration backup

1. **System → Settings → Logging → Remote → +**: Enabled ✓, Transport **UDP(4)**, Applications empty (= all), Hostname `10.2.0.102`, Port `514`, rfc5424 ✗, Description *Wazuh SIEM01* → Save → Apply. (Wazuh side: chapter 11.5.)
2. **System → Configuration → Backups → Download**: tick **Encrypt this configuration file**, password, keep “Do not backup RRD data” → **Download configuration**. Copy the XML to `\\nas01\Sauvegardes\OPNSENSE`. Repeat after every change.

Part 3 — Network, track B: real Cisco hardware

Router-on-a-stick, managed switch and Proxmox server. Same IP plan and same rules as track A, translated into IOS commands.

1B. From VMware to hardware

In the VMware lab, OPNsense was router, firewall and NAT, and each VMware network was a VLAN. On hardware, the same jobs are split between devices:

Function	VMware lab	Hardware version
Zones / VLANs	VMnet1, LAN segments SRV20, LAN10, DMZ40	VLANs 30, 20, 10, 40 on the switch
Carrying several zones on one cable	—	802.1Q trunks (switch ↔ router, switch ↔ Proxmox)
Routing between zones	OPNsense interfaces em1–em4	router sub-interfaces G0/0.10/.20/.30/.40
Filtering	OPNsense rules (stateful)	extended ACLs (stateless: replies allowed with *established*)
DHCP relay	Services → DHCRelay	`ip helper-address 10.2.0.70`
Internet access	NAT on the WAN	NAT overload (PAT) on G0/1
GLPI publication	Destination NAT 443 → PROXY01	static PAT 443 → 10.2.0.114
Logs to the SIEM	remote syslog	`logging host 10.2.0.102`
Admin access	GUI on ADMINISTRATION only	SSH only, vty restricted to 10.2.0.96/28
Hypervisor	VMware Workstation	Proxmox, VM NICs tagged on a VLAN-aware bridge

The hardware topology is Figure 3 in Part 0.

2B. Equipment, cabling and plan

2B.1 Shopping list (used gear is fine)

Item	Model examples	Note
Router	Cisco ISR 1941 or 2901 (IOS 15.x)	avoid 1841 (IOS 12.4); 2 GE ports needed
Switch	Catalyst 2960 / 2960-Plus / 2960-X, or fanless 2960-8TC-L / 2960C	48-port models are loud
Console cable	USB → RJ45 (FTDI chip)	mandatory for the first configuration
Hypervisor	PC or mini PC with Proxmox (e.g. Lenovo M920q, 32 GB)	a 2nd disk for backups
Cables	RJ45 patch cables (+1 crossover for very old ports)	
Optional	Catalyst 3560 (layer 3)	inter-VLAN routing with SVIs (Appendix B)

2B.2 Addressing plan

Zone	VLAN	Network	Gateway (router)	Static devices
UTILISATEURS	10	10.2.0.0/26	10.2.0.1	DHCP .10–.60 from SRV-AD
SERVEURS	20	10.2.0.64/27	10.2.0.65	SRV-AD .70 · FS01 .71 · LNX01 .72 · NAS01 .73
ADMINISTRATION	30	10.2.0.96/28	10.2.0.97	switch .98 · Proxmox .100 · admin PCs .101–.108 · SIEM01 .102
DMZ	40	10.2.0.112/28	10.2.0.113	PROXY01 .114
Parking	999	—	—	unused ports and native VLAN of trunks
WAN	—	ISP box LAN (e.g. 192.168.1.0/24)	ISP box	router G0/1 by DHCP

Note — SIEM01 moves to the **ADMINISTRATION** VLAN as in the VMware lab; the admin PC overlaps it at .101–.108, so give your admin PCs .101, .103–.108 and keep .102 for SIEM01.

2B.3 Port and cabling table

Device	Port	Mode	VLAN	Connected to
Router	G0/1	routed (outside)	—	ISP box
Router	G0/0	trunk (sub-interfaces)	10, 20, 30, 40	switch Gi0/1
Switch	Gi0/1	trunk	10, 20, 30, 40 (native 999)	router G0/0
Switch	Gi0/2	trunk	10, 20, 30, 40 (native 999)	Proxmox NIC
Switch	Fa0/1–4	access	30	admin PCs
Switch	Fa0/5–8	access	10	user PCs
Switch	Fa0/9–24	access, shutdown	999	unused
Admin PC	NIC	—	30	static 10.2.0.101/28 · gw 10.2.0.97 · DNS 10.2.0.70

Why — The admin PC now has a real gateway (10.2.0.97): no persistent route is needed any more, unlike the VMware lab where the host adapter had no gateway.

3B. Console access and factory reset

3B.1 Connect

1. Plug the console cable into the **CONSOLE** port (blue/RJ45) and a USB port of the admin PC.
2. Windows: **Gestionnaire de périphériques → Ports (COM et LPT)** → note the COM number.
3. PuTTY (or Tera Term): Serial, COMx, **9600** baud, 8 data bits, no parity, 1 stop bit, flow control none.
4. Press Enter. Answer **no** to “Would you like to enter the initial configuration dialog?”.

3B.2 Start from a clean configuration (used equipment)

```
! ---- switch
enable
delete flash:vlan.dat          (confirm with Enter)
erase startup-config
reload                         (answer "no" if asked to save)

! ---- router
enable
erase startup-config
reload
```

Watch out — On a switch, VLANs are stored in **vlan.dat**, not in the configuration: erasing only the startup-config keeps the old VLANs. If an enable password blocks you, use the password-recovery procedure of the model (break sequence / MODE button) — good practice for the CCNA anyway.

4B. Baseline: identity, passwords, SSH

The same base on both devices (shown for the switch; change the hostname to **R-LAB** on the router and use `line vty 0 4`).

```
enable
configure terminal
hostname SW-LAB
no ip domain-lookup
ip domain-name lab.local
enable secret <enable-secret>
service password-encryption
username admin privilege 15 secret <admin-secret>
banner motd # Acces reserve aux administrateurs du lab #
!
line console 0
 login local
 logging synchronous
 exec-timeout 10 0
!
crypto key generate rsa modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
!
access-list 30 remark Admin network only
access-list 30 permit 10.2.0.96 0.0.0.15
line vty 0 15
 transport input ssh
 login local
 access-class 30 in
 exec-timeout 10 0
end
copy running-config startup-config
```

Command	Why
no ip domain-lookup	a typo is not sent to a DNS lookup that freezes the console
enable secret / username … secret	hashed secrets (type 9/5) instead of reversible passwords
crypto key … / ip ssh version 2	SSH needs a hostname, a domain name and an RSA key
transport input ssh	Telnet (clear text) refused
access-class 30 in	SSH only from 10.2.0.96/28 — replaces the OPNsense rule “GUI on ADMINISTRATION only”

Check — `show ip ssh` → SSH Enabled - version 2.0. Later, from the admin PC: `ssh admin@10.2.0.98` (switch) and `ssh admin@10.2.0.97` (router).

5B. Switch: VLANs, access ports, trunks

5B.1 VLANs

```
configure terminal
vlan 10
 name UTILISATEURS
vlan 20
 name SERVEURS
vlan 30
 name ADMINISTRATION
vlan 40
 name DMZ
vlan 999
 name PARKING
exit
```

5B.2 Access ports

```
spanning-tree mode rapid-pvst
!
interface range fa0/1 - 4
 description PC-ADMIN
 switchport mode access
 switchport access vlan 30
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface range fa0/5 - 8
 description PC-UTILISATEUR
 switchport mode access
 switchport access vlan 10
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
!
interface range fa0/9 - 24
 description NON-UTILISE
 switchport mode access
 switchport access vlan 999
 shutdown
```

Why — portfast: an access port forwards immediately (no 30 s STP wait at boot, DHCP works). **bpduguard**: if someone plugs a switch there, the port goes err-disabled instead of creating a loop. **port-security** limits the MAC addresses on user ports. Unused ports are shut and parked in VLAN 999.

5B.3 Trunks

```
interface gi0/1
 description TRUNK-ROUTEUR
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
!
interface gi0/2
 description TRUNK-PROXMOX
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
```

Watch out — On a layer-3 switch (3560/3650) add `switchport trunk encapsulation dot1q` **before** `switchport mode trunk`. The native VLAN must be the same at both ends of a trunk (CDP reports “native VLAN mismatch”); 999 carries nothing, which avoids VLAN-hopping tricks.

5B.4 Management address

```
interface vlan 30
 description MANAGEMENT
 ip address 10.2.0.98 255.255.255.240
 no shutdown
interface vlan 1
 shutdown
ip default-gateway 10.2.0.97
end
copy running-config startup-config
```

Check — `show vlan brief` (ports in the right VLANs) · `show interfaces trunk` (Gi0/1, Gi0/2 trunking, allowed 10,20,30,40, native 999) · `show interfaces status` · `show port-security`.

6B. Router: router-on-a-stick and DHCP relay

```
configure terminal
interface g0/0
 description TRUNK-SWITCH
 no ip address
 no shutdown
!
interface g0/0.10
 description VLAN10-UTILISATEURS
 encapsulation dot1Q 10
 ip address 10.2.0.1 255.255.255.192
 ip helper-address 10.2.0.70
!
interface g0/0.20
 description VLAN20-SERVEURS
 encapsulation dot1Q 20
 ip address 10.2.0.65 255.255.255.224
!
interface g0/0.30
 description VLAN30-ADMINISTRATION
 encapsulation dot1Q 30
 ip address 10.2.0.97 255.255.255.240
!
interface g0/0.40
 description VLAN40-DMZ
 encapsulation dot1Q 40
 ip address 10.2.0.113 255.255.255.240
!
interface g0/1
 description WAN-BOX
 ip address dhcp
 no shutdown
end
```

Why — One physical port, four logical gateways: each sub-interface handles the frames tagged with its VLAN. `ip helper-address` turns the DHCP broadcasts of VLAN 10 into unicast packets to SRV-AD (option 003 of the scope must be 10.2.0.1). `ip address dhcp` on G0/1 also installs the default route given by the ISP box.

Check — `show ip interface brief` (all up/up) · `show ip route` (four connected networks + default route) · from the admin PC: ping 10.2.0.1, .65, .97, .113 · a user PC gets 10.2.0.10–.60 with DHCP server 10.2.0.70.

7B. NAT and GLPI publication

```
configure terminal
interface g0/0.10
 ip nat inside
interface g0/0.20
 ip nat inside
interface g0/0.30
 ip nat inside
interface g0/0.40
 ip nat inside
interface g0/1
 ip nat outside
!
ip access-list standard NAT-LAB
 permit 10.2.0.0 0.0.0.255
ip nat inside source list NAT-LAB interface g0/1 overload
!
ip nat inside source static tcp 10.2.0.114 443 interface g0/1 443
end
```

Line	Role
ip nat inside / outside	which side is the lab, which side is the Internet (swapping them = no NAT at all)
… list NAT-LAB interface g0/1 overload	PAT: the whole lab goes out with the WAN address (equivalent of OPNsense's automatic outbound NAT)
static tcp 10.2.0.114 443 interface g0/1 443	port forwarding: WAN:443 → PROXY01:443 (Destination NAT)

Note — To test the publication, browse the router's WAN address from a PC on the ISP box network. On a real Internet line, also forward 443 on the ISP box to the router (or put the router in the box's DMZ). The GUI/HTTP server of the router stays off: `no ip http server` and `no ip http secure-server`.

Check — `show ip nat translations` · `show ip nat statistics` · from a server: ping 1.1.1.1.

8B. ACLs — the firewall rules in IOS

Extended ACLs are applied **inbound** on the sub-interface of the zone they protect against. Unlike OPNsense, an ACL is **stateless**: it does not remember connections. Replies to TCP sessions opened from elsewhere are allowed with `established`; replies to traffic leaving the zone are not filtered because the ACL only checks packets entering the router from that zone.

8B.1 UTILISATEURS (VLAN 10)

```
ip access-list extended ACL-USERS
 remark --- DHCP to the relay
 permit udp any any eq bootps
 remark --- replies to sessions opened from other zones
 permit tcp 10.2.0.0 0.0.0.63 any established
 permit icmp 10.2.0.0 0.0.0.63 any echo-reply
 remark --- Wazuh agents to SIEM01 only
 permit tcp 10.2.0.0 0.0.0.63 host 10.2.0.102 range 1514 1515
 remark --- admin network and router management are forbidden
 deny   ip 10.2.0.0 0.0.0.63 10.2.0.96 0.0.0.15 log
 deny   tcp 10.2.0.0 0.0.0.63 host 10.2.0.1 eq 22 log
 remark --- everything else (servers, Internet)
 permit ip 10.2.0.0 0.0.0.63 any
!
interface g0/0.10
 ip access-group ACL-USERS in
```

8B.2 DMZ (VLAN 40)

```
ip access-list extended ACL-DMZ
 permit tcp any any established
 permit udp host 10.2.0.114 host 10.2.0.70 eq domain
 permit tcp host 10.2.0.114 host 10.2.0.70 eq domain
 permit tcp host 10.2.0.114 host 10.2.0.72 eq www
 permit tcp host 10.2.0.114 host 10.2.0.102 range 1514 1515
 deny   ip any 10.2.0.0 0.0.0.255 log
 permit tcp host 10.2.0.114 any eq www
 permit tcp host 10.2.0.114 any eq 443
 permit udp host 10.2.0.114 any eq ntp
 deny   ip any any log
!
interface g0/0.40
 ip access-group ACL-DMZ in
```

Why — Same logic as the OPNsense DMZ rules: DNS, GLPI:80 and the SIEM agent ports are the only ways into the lab; the deny to 10.2.0.0/24 (logged) comes **before** the Internet permits; the final deny is implicit anyway but adding `log` makes it visible.

8B.3 SERVEURS and ADMINISTRATION

No ACL, as in the VMware lab (“Serveurs vers tout”, “Default allow LAN”). Optional hardening: restrict servers' outbound Internet to 80/443/123/53 with an ACL-SERVERS built the same way.

8B.4 Editing an ACL safely

```
show access-lists ACL-USERS          (each line has a sequence number: 10, 20, 30…)
configure terminal
ip access-list extended ACL-USERS
 15 permit tcp 10.2.0.0 0.0.0.63 host 10.2.0.71 eq 445    (inserted between 10 and 20)
 no 40                                                    (removes line 40)
```

Watch out — Order matters and the end of every ACL is an invisible **deny any**. Apply a new ACL from the console (not over SSH) or schedule a safety reload: `reload in 10` before the change, `reload cancel` once it works.

Check — `show access-lists` (match counters increase) · `show ip interface g0/0.10` (“Inbound access list is ACL-USERS”) · from a user PC: ping 10.2.0.97 fails, `Test-NetConnection 10.2.0.102 -Port 1514` True, -Port 443 False.

9B. Time, logs to Wazuh, configuration backups

9B.1 NTP and timestamps (both devices)

```
configure terminal
clock timezone CET 1 0
clock summer-time CEST recurring last Sun Mar 2:00 last Sun Oct 3:00
ntp server 10.2.0.70
service timestamps log datetime msec localtime show-timezone
service timestamps debug datetime msec localtime
end
show ntp associations          (* in front of 10.2.0.70 = synchronised)
```

9B.2 Syslog to SIEM01

```
configure terminal
logging host 10.2.0.102
logging trap informational
logging source-interface g0/0.30        (router; on the switch: vlan 30)
end
```

On SIEM01, allow the two devices in the syslog block of `/var/ossec/etc/ossec.conf`, then restart wazuh-manager:

```
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>10.2.0.97</allowed-ips>
  <allowed-ips>10.2.0.98</allowed-ips>
  <local_ip>10.2.0.102</local_ip>
</remote>
```

Note — Wazuh ships decoders and rules for Cisco IOS messages (%SEC-6-IPACCESSLOGP for logged ACL hits, %SYS-5-CONFIG_I for configuration changes, login failures...). Check what your messages produce with `wazuh-logtest`, exactly as for OPNsense.

9B.3 Saving configurations

Method	Command
Local	copy running-config startup-config (after every change)
TFTP to the admin PC (Tftpd64)	copy running-config tftp: → 10.2.0.101 → R-LAB.cfg
Manual	terminal length 0 · show running-config → paste into R-LAB.txt / SW-LAB.txt on the NAS

Keep dated copies in `\\nas01\Sauvegardes\CISCO` (equivalent of the OPNsense XML).

10B. Proxmox on the trunk

10B.1 Management address in VLAN 30

Install Proxmox, then edit `/etc/network/interfaces` from the physical console (keep a copy: `cp /etc/network/interfaces /root/interfaces.bak`):

```
auto lo
iface lo inet loopback

iface eno1 inet manual

auto vmbr0
iface vmbr0 inet manual
    bridge-ports eno1
    bridge-stp off
    bridge-fd 0
    bridge-vlan-aware yes
    bridge-vids 10 20 30 40

auto vmbr0.30
iface vmbr0.30 inet static
    address 10.2.0.100/28
    gateway 10.2.0.97
```

`ifreload -a`, then from the admin PC: https://10.2.0.100:8006. Rollback: restore the .bak file and `ifreload -a`.

10B.2 VM network cards

VM	Bridge	VLAN Tag	IP (unchanged)
SRV-AD, FS01, LNX01, NAS01, CA01	vmbr0	20	.70 / .71 / .72 / .73 / .74
SIEM01	vmbr0	30	.102
PROXY01	vmbr0	40	.114
CLT01 (test VM)	vmbr0	10	DHCP

> **Note —** Windows sees a new network card when a VM moves from VMware to Proxmox (VirtIO or e1000): set its static IP again inside the VM. VirtIO cards need the virtio-win drivers.

10B.3 Snapshots and backups

- Snapshot **BASE-PROPRE** of each VM right after OS install and updates.
- Third disk → **Datacenter → Storage → Add → Directory**, content VZDump backup file.
- **Datacenter → Backup → Add**: all VMs, mode Snapshot, ZSTD, retention 3, nightly.
- Restore test to a **new VMID** with **Unique** (new MAC) while the original is stopped — never a copy of the DC while the original runs.

11B. Tests

#	Test	From	Expected
1	show interfaces trunk	switch	Gi0/1, Gi0/2 trunking, VLANs 10,20,30,40, native 999
2	show ip interface brief	router	G0/0.x and G0/1 up/up
3	ping the 4 gateways	admin PC	all reply
4	ipconfig /all	user PC	10.2.0.10–.60, gw 10.2.0.1, DHCP server 10.2.0.70
5	ping srv-ad.lab.local / 1.1.1.1	user PC	reply / reply
6	ping 10.2.0.97, ssh admin@10.2.0.1	user PC	fail (ACL hits visible in show access-lists)
7	Test-NetConnection 10.2.0.102 -Port 1514 / 443	user PC	True / False
8	ssh admin@10.2.0.97 / .98	admin PC	OK
9	ssh from a user PC	user PC	refused (access-class)
10	PROXY01 → 10.2.0.70:445 and 10.2.0.72:22	PROXY01	blocked, logged
11	https://<router WAN IP>	PC on ISP network	GLPI login page through PROXY01
12	show ip nat translations	router	PAT entries + static 443
13	show ntp associations	both	synchronised to 10.2.0.70
14	Wazuh Threat Hunting, location 10.2.0.97	admin PC	Cisco events (ACL hits, config changes)
15	plug a small switch on Fa0/5	—	port err-disabled (bpduguard)

12B. Troubleshooting

Symptom	Likely cause	Check / fix
Users get 169.254.x.x	no helper-address, ACL blocks bootps, scope inactive, port in wrong VLAN	show run int g0/0.10 · show vlan brief · DHCP console on SRV-AD
A VLAN cannot reach its gateway	VLAN not allowed on the trunk / wrong dot1Q number	show interfaces trunk · show run int g0/0.x
CDP: native VLAN mismatch	native VLAN different at each end	same native vlan on both trunk ends
Trunk does not come up	DTP disabled on one side only, encapsulation missing (L3 switch)	switchport mode trunk on both, trunk encapsulation dot1q
Port down “err-disabled”	bpduguard or port-security violation	show interfaces status err-disabled · fix the cause · shutdown / no shutdown
No Internet from the lab	ip nat inside/outside swapped, NAT ACL wrong, no default route	show ip nat statistics · show ip route
Everything blocked after applying an ACL	implicit deny, missing established/return traffic	show access-lists counters · reload in 10 before testing
SSH refused from the admin PC	no RSA key, access-class, transport input	show ip ssh · show run | section line vty
No logs in Wazuh	allowed-ips missing, wrong source interface, ACL	tcpdump -ni any udp port 514 on SIEM01
Wrong time in logs	NTP not synchronised, timezone missing	show clock detail · show ntp status
Proxmox unreachable after network change	VLAN 30 not on the trunk, wrong vmbr0.30 config	console: ip a, cat /etc/network/interfaces, restore .bak

Part 4 — Servers and services (both tracks)

Active Directory, file server, GPOs, client, GLPI, NAS, backups, SIEM, DMZ. Identical for both tracks; only the way a VM is plugged into its VLAN changes.

The chapters are written with VMware words. On track B (Proxmox), translate them with this table:

Track A (VMware Workstation)	Track B (Proxmox + Cisco)
Card on **segment SRV20**	vmbr0, **VLAN tag 20**
Card on **VMnet1** (admin)	vmbr0, **VLAN tag 30**
Card on **segment LAN10**	vmbr0, **VLAN tag 10**
Card on **segment DMZ40**	vmbr0, **VLAN tag 40**
VMware Tools / open-vm-tools	QEMU guest agent (VM → Options → Enabled; Windows: virtio-win ISO; Linux: qemu-guest-agent)
Disk / NIC type	VirtIO SCSI + VirtIO net (load virtio-win drivers during Windows setup) or SATA + e1000
VMware snapshot	Proxmox snapshot (take **BASE-PROPRE** right after OS install + updates)
“OPNsense rule X” / alias	matching ACL line in 8B
OPNsense DHCP relay	ip helper-address (6B)
OPNsense remote syslog	logging host (9B)
Destination NAT 443	ip nat inside source static tcp … 443 (7B)
Host PC route 10.2.0.0/25	not needed: the admin PC has gateway 10.2.0.97
Gateway .65 / .97 / .1 / .113 = OPNsense	same addresses = router sub-interfaces

Note — Gateways, IPs, DNS, accounts and every server setting are the same in both tracks — that is the point of keeping one IP plan.

3. SRV-AD — domain controller, DNS, DHCP, time

The heart of the lab. **Golden rule: static IP and final name before promoting the server.**

3.1 VM and installation

Field	Value
VM	02 - SRV-AD · Windows Server 2022 · 2 vCPU · 3–4 GB · 60 GB · segment **SRV20**
Edition	Windows Server 2022 Standard Evaluation **(expérience de bureau)**
After install	VMware Tools → reboot · Windows Update

3.2 Name and IP

1. **Gestionnaire de serveur → Serveur local** → computer name → **Modifier…** → `SRV-AD` → reboot.
2. Ethernet0 → **TCP/IPv4**: IP **10.2.0.70**, mask **255.255.255.224**, gateway **10.2.0.65**, DNS **127.0.0.1** (use 1.1.1.1 temporarily if you run Windows Update before promotion).

Check — `ping 10.2.0.65` and `ping 1.1.1.1` answer (SERVEURS rule + NAT).

3.3 Roles and promotion

1. **Gérer → Ajouter des rôles et fonctionnalités**: **AD DS**, **Serveur DNS**, **Serveur DHCP**.
2. Flag → **Promouvoir ce serveur en contrôleur de domaine → Ajouter une nouvelle forêt →** `lab.local` (NetBIOS LAB) → DSRM password → install → reboot.
3. Check DNS is **127.0.0.1** after promotion.

```
dcdiag /q                    (no output = healthy)
nltest /dsgetdc:lab.local    (ends with "La commande a réussi")
nslookup lab.local           (answers 10.2.0.70)
```

Note — .local is also used by mDNS (Bonjour). It is fine for a Windows lab; for a new real domain prefer a sub-domain of a name you own (e.g. ad.example.com) or a reserved name (lab.home.arpa).

3.4 DNS: forwarders and reverse zone

1. **Outils → DNS** → right-click **SRV-AD → Propriétés → Redirecteurs**: `1.1.1.1`, `8.8.8.8`.
2. **Zones de recherche inversée → Nouvelle zone**: Zone principale, stored in AD, to all DNS servers of the domain, IPv4, network ID `10.2.0`, secure dynamic updates only.
3. Later, for each Linux server, **lab.local → Nouvel hôte (A)** with ✓ **Créer un pointeur PTR associé** (lnx01 .72, nas01 .73, siem01 .102, proxy01 .114). Windows members register themselves.

Check — `nslookup 10.2.0.72 10.2.0.70` → lnx01.lab.local. On the DC itself nslookup may show “Serveur: UnKnown / ::1”: that is IPv6 localhost without PTR, harmless.

3.5 Time source (NTP)

The DC holding the PDC emulator role is the time source of the domain. Without configuration it follows its own clock (“Local CMOS Clock”) and Linux clients refuse it.

```
w32tm /config /manualpeerlist:"0.fr.pool.ntp.org 1.fr.pool.ntp.org" /syncfromflags:manual
/reliable:yes /update
net stop w32time
net start w32time
w32tm /resync
w32tm /query /status          (Source: x.fr.pool.ntp.org · Couche 2–4)
```

Why — Kerberos refuses logons when clocks differ by more than 5 minutes, and the SIEM needs one common time to put events from different machines in the right order.

3.6 DHCP

1. **Outils → DHCP** → right-click the server → **Autoriser** → F5 (IPv4 turns green).
2. **IPv4 → Nouvelle étendue…**:

Field	Value
Nom	VLAN10-Utilisateurs
Plage	10.2.0.10 → 10.2.0.60, longueur 26 (255.255.255.192)
Exclusions / bail	none / 8 jours
003 Routeur	10.2.0.1
006 DNS / 015 Domaine	10.2.0.70 / lab.local (remove any other DNS)
WINS	none
Activer	Oui

Watch out — A DC must never get its own IP by DHCP: event 1041 (DHCP refuses to serve) and the firewall profile stuck on “Privé” instead of “Domaine” are typical symptoms.

4. Active Directory structure (AGDLP)

4.1 Organisational units

```
lab.local
 ├─ COMPTES-ADMINISTRATION   (SVC-Backup, svc-glpi, admin accounts)
 ├─ GROUPES                  (GG_* and DL_FS_*)
 ├─ ORDINATEURS              (CLT01)
 ├─ SERVEURS                 (FS01 — SRV-AD stays in Domain Controllers)
 └─ UTILISATEURS
     ├─ COMMERCIAL  ├─ COMPTABILITE  ├─ DIRECTION
     ├─ INFORMATIQUE └─ TECHNIQUE
```

dsa.msc → right-click the domain (or UTILISATEURS) → **Nouveau → Unité d'organisation**. To move an OU later: **Affichage → Fonctionnalités avancées** → OU → Propriétés → **Objet** → untick **Protéger l'objet des suppressions accidentelles** → Déplacer.

4.2 Users

Nouveau → Utilisateur in the service OU. Logon name **prenom.nom** without accents; password ≥ 12 characters with complexity; untick “doit changer” for the lab.

OU	Users (logon names)	Global group
DIRECTION	alice.martin, hugo.bernard	GG_DIRECTION
COMPTABILITE	emma.laurent, lea.robert, nina.lopez	GG_COMPTABILITE
COMMERCIAL	jean.boulet, ines.durand, lucas.morel	GG_COMMERCIAL
TECHNIQUE	sarah.petit, noah.leroy, yasmine.fontaine	GG_TECHNIQUE
INFORMATIQUE	maxime.roux, clara.garnier, test.recette	GG_INFORMATIQUE (not test.recette)
COMPTES-ADMINISTRATION	SVC-Backup (member of **Opérateurs de sauvegarde**), svc-glpi — both “mot de passe n'expire jamais”	—

Optional PowerShell for one account (password asked in hidden mode):

```
$mdp = Read-Host -AsSecureString "Mot de passe"
New-ADUser -Name "Test Recette" -GivenName Test -Surname Recette -SamAccountName test.recette `
  -UserPrincipalName test.recette@lab.local `
  -Path "OU=INFORMATIQUE,OU=UTILISATEURS,DC=lab,DC=local" `
  -AccountPassword $mdp -Enabled $true -ChangePasswordAtLogon $false
```

4.3 Groups and nesting

In **GROUPES**: 5 global groups (scope **Globale**) and 9 domain local groups (scope **Domaine local** — the dialog defaults to Globale). Nest from each GG: **Membre de → Ajouter**, names separated by ;, **Vérifier les noms**.

Global group (members = users)	Member of (domain local groups)
GG_DIRECTION	DL_FS_COMMUN_RW; DL_FS_DIRECTION_RW; DL_FS_COMPTABILITE_**R**; DL_FS_COMMERCIAL_**R**; DL_FS_TECHNIQUE_**R**
GG_COMPTABILITE	DL_FS_COMMUN_RW; DL_FS_COMPTABILITE_RW
GG_COMMERCIAL	DL_FS_COMMUN_RW; DL_FS_COMMERCIAL_RW
GG_TECHNIQUE	DL_FS_COMMUN_RW; DL_FS_TECHNIQUE_RW
GG_INFORMATIQUE	DL_FS_COMMUN_RW; DL_FS_INFORMATIQUE_RW

Why — AGDLP: Accounts → Global groups (who: the service) → Domain Local groups (what: one resource + one access level) → Permissions given **only to DL groups**. Moving a person = changing one membership, never editing folder permissions.

Check — DL_FS_COMMUN_RW → Membres lists the 5 GG. Type column: 9 × “Domaine local”, 5 × “Global”.

5. FS01 — file server

5.1 VM, name, IP, domain

Field	Value
VM	05 - FS01 · Windows Server 2025 Datacenter Evaluation (expérience de bureau) · 2 vCPU · 3–4 GB · segment SRV20
Disks	60 GB system + **40 GB data added at creation** (same bus type)
Name / IP	FS01 · 10.2.0.71/27 · gw 10.2.0.65 · DNS **10.2.0.70**

1. VMware Tools → rename FS01 → reboot → static IP.
2. Test `nslookup lab.local` (→ 10.2.0.70), then **sysdm.cpl → Modifier → Domaine** `lab.local` with LAB\Administrateur → reboot.
3. Log on as LAB\`Administrateur` (local admin: `.\Administrateur`).
4. On SRV-AD, move the computer object FS01 into OU **SERVEURS**.

5.2 Open file sharing in the Windows firewall

Windows Server 2025 keeps the *File and Printer Sharing* rules disabled: FS01 answers neither ping nor SMB. PowerShell as admin (language-independent group ID):

```
Get-NetConnectionProfile                          (NetworkCategory: DomainAuthenticated)
Enable-NetFirewallRule -Group "@FirewallAPI.dll,-28502"
```

Check — From CLT01 or SRV-AD: `Test-NetConnection 10.2.0.71 -Port 445` → TcpTestSucceeded True.

5.3 Data disk

1. **Gestion des disques** → initialise the 40 GB disk **GPT**.
2. If the CD-ROM holds D:, move it to **E:** (right-click → Modifier la lettre de lecteur).
3. **Nouveau volume simple** → letter **D**, NTFS, name **DATA**.
4. Create `D:\PARTAGES\` with COMMUN, DIRECTION, COMPTABILITE, COMMERCIAL, TECHNIQUE, INFORMATIQUE.

5.4 NTFS permissions and shares

For each folder: **Sécurité → Avancé → Désactiver l'héritage → Convertir**, remove **Utilisateurs**, keep Système / Administrateurs / CREATEUR PROPRIETAIRE, add the DL group(s). Then **Partage avancé → Partager ce dossier → Autorisations**: remove **Tout le monde**, add the same DL group(s).

Share	DL group	NTFS	Share permission
COMMUN	DL_FS_COMMUN_RW	Modification	Modifier + Lecture
DIRECTION	DL_FS_DIRECTION_RW	Modification	Modifier + Lecture
COMPTABILITE	DL_FS_COMPTABILITE_RW / _R	Modification / Lecture et exécution	Modifier / Lecture
COMMERCIAL	DL_FS_COMMERCIAL_RW / _R	Modification / Lecture et exécution	Modifier / Lecture
TECHNIQUE	DL_FS_TECHNIQUE_RW / _R	Modification / Lecture et exécution	Modifier / Lecture
INFORMATIQUE	DL_FS_INFORMATIQUE_RW	Modification	Modifier + Lecture

Watch out — RW = **Modification**, never Contrôle total (users could change permissions). Over the network the **most restrictive** of share and NTFS applies. Check that **Emplacements** = lab.local when adding groups.

Why — LAB\Administrateur gets “accès refusé” on \\FS01\Commun: correct, it is in no DL group. Admins work locally or through `\\FS01\D$\PARTAGES`.

5.5 Wallpaper folder

1. In `D:\PARTAGES\COMMUN` create **FondEcran**, copy an image as `novatech.jpg` (show extensions to avoid novatech.jpg.jpg).
2. FondEcran → disable inheritance (Convertir) → DL_FS_COMMUN_RW: **read only** (untick Modification, Écriture).

6. GPOs and logon hours

Outils → Gestion de stratégie de groupe. Names are imposed by the project.

6.1 GPO-DOMAINE-Politique-Comptes (domain root)

1. Right-click **lab.local → Créer un objet GPO dans ce domaine, et le lier ici**.
2. Configuration ordinateur → Stratégies → Paramètres Windows → Paramètres de sécurité → Stratégies de comptes:

Policy	Value
Historique des mots de passe	5
Longueur minimale	12
Complexité	Activé
Seuil de verrouillage (set first)	5 tentatives
Durée de verrouillage	15 min
Réinitialiser le compteur après	15 min

3. On the domain, tab **Objets de stratégie de groupe liés**: move it to **link order 1** (above Default Domain Policy).

Check — `gpupdate /force` then `net accounts` → 12 / 5 / 5 / 15 / 15.

6.2 GPO-UTIL-Fond-Ecran (OU UTILISATEURS)

User Configuration → Stratégies → Modèles d'administration → **Bureau → Bureau → Papier peint du Bureau**: Activé, `\\FS01\Commun\FondEcran\novatech.jpg`, style Remplir.

Watch out — Use the **share** path (not D:\, not D$, not \\FS01\PARTAGES which is a folder), file name with its extension, no accents.

6.3 GPO-UTIL-Lecteurs (OU UTILISATEURS)

Configuration utilisateur → **Préférences** → Paramètres Windows → **Mappages de lecteurs** → Nouveau → Lecteur mappé: Action **Mettre à jour**, Reconnecter ✓. Tab **Commun → Ciblage au niveau de l'élément** → Groupe de sécurité; for two groups select the second line → **Options d'élément → Ou**.

Letter	Location	Targeting
P:	\\FS01\Commun	none (everyone)
D:	\\FS01\Direction	GG_DIRECTION
F:	\\FS01\Comptabilite	GG_COMPTABILITE **OU** GG_DIRECTION
V:	\\FS01\Commercial	GG_COMMERCIAL **OU** GG_DIRECTION
T:	\\FS01\Technique	GG_TECHNIQUE **OU** GG_DIRECTION
I:	\\FS01\Informatique	GG_INFORMATIQUE

Why — The GPO only **shows** drives; access rights still come from AGDLP (Direction sees F:, V:, T: read-only). Targeting with **ET** instead of **OU** gives the drive to nobody.

6.4 GPO-METIERS-Blocage-CMD (4 business OUs)

1. Create it under **Objets de stratégie de groupe** (not linked).
2. User Configuration → Stratégies → Modèles d'administration → Système → **Désactiver l'accès à l'invite de commandes** → Activé, script processing **Non**.
3. **Lier un objet GPO existant** on DIRECTION, COMPTABILITE, COMMERCIAL, TECHNIQUE — never INFORMATIQUE, UTILISATEURS or COMPTES-ADMINISTRATION.

Check — GPO → tab **Étendue**: exactly 4 links.

6.5 Logon hours

1. In each business OU (DIRECTION, COMPTABILITE, COMMERCIAL, TECHNIQUE) select all users → **Propriétés → Compte** → tick **Horaires d'accès** → open it.
2. **Tout → Connexion refusée**, then drag lundi 8:00 → vendredi 16:00–17:00 → **Connexion autorisée**.
3. Skip INFORMATIQUE and admin/service accounts.

Note — AD also enforces these hours for **LDAP logons**: business users cannot log into GLPI outside Mon–Fri 08:00–17:00 either. An open session is not closed at 17:00.

7. CLT01 — client workstation

Field	Value
VM	08 - CLT01 · Windows 11 **Pro** (Home cannot join a domain) · 2 vCPU · 3–4 GB · 64 GB · segment **LAN10**
TPM	VMware requires encryption: “Seuls les fichiers nécessaires à la prise en charge d'un TPM sont chiffrés” — keep that password
Local account	Pro: at the account screen **Shift + F10** → `start ms-cxh:localonly`
Network	automatic (DHCP)

1. Install VMware Tools, reboot, then `ipconfig /all`.
2. Rename `CLT01` and join `lab.local` in one go (sysdm.cpl) → reboot.
3. On SRV-AD move CLT01 from **Computers** to **ORDINATEURS**.

Check — ipconfig: 10.2.0.10–.60 · mask 255.255.255.192 · gateway 10.2.0.1 · **Serveur DHCP 10.2.0.70** (proves the relay) · DNS 10.2.0.70 · suffix lab.local. Lease visible in DHCP → Baux d'adresses.

Tests from CLT01

Test	Expected
ping srv-ad.lab.local · ping 1.1.1.1	replies (users → servers and Internet allowed)
https://10.2.0.1 in Edge	does not load (firewall blocked)
Test-NetConnection 10.2.0.102 -Port 1514 / 443	True / False (narrow SIEM exception)
emma.laurent → \\FS01	COMMUN, COMPTABILITE OK; DIRECTION “accès refusé”
alice.martin → \\FS01\COMPTABILITE	opens; creating a file refused (read-only)
Drives after logon	Direction P D F V T · Compta P F · Commercial P V · Technique P T · Info P I
Wallpaper	applied after logoff/logon (a restart may be needed the first time)
cmd as emma.laurent / maxime.roux	blocked / allowed; gpresult /r shows the GPO list
Logon outside hours (business account)	refused

8. LNX01 — GLPI server

A Debian server (no desktop) running Apache, PHP and MariaDB for GLPI, linked to Active Directory, administered by SSH key only.

8.1 VM and Debian installation

Field	Value
VM	06 - LNX01 · Debian 13 (netinst ISO) · 2 vCPU · 2 GB · 20 GB · segment SRV20
Installer network	DHCP fails (none in SRV20) → **Configurer vous-même le réseau**: 10.2.0.72/27 · gw 10.2.0.65 · DNS 10.2.0.70
Hostname / domain	lnx01 / lab.local
Root password	**leave empty** → sudo is installed and given to your user
User	e.g. lnxadmin (admin, sudo)
Disk	assisted, whole disk, single partition
Software	✗ desktop environment · ✓ **SSH server** · ✓ standard system utilities

1. After reboot: `sudo apt update && sudo apt full-upgrade -y` and `sudo apt install -y open-vm-tools`.
2. On SRV-AD: A record **lnx01 → 10.2.0.72** with PTR.
3. From the host: `ping 10.2.0.72`.

8.2 Web stack

```
sudo apt install -y apache2 mariadb-server php libapache2-mod-php php-mysql php-curl php-gd \
  php-intl php-xml php-mbstring php-bcmath php-zip php-bz2 php-ldap php-opcache php-apcu
sudo mariadb-secure-installation
```

> **Note —** GLPI 11 requires PHP ≥ 8.2 and MariaDB ≥ 10.6; Debian 13 ships newer versions of both. Check the release notes of the GLPI version you install for its exact requirements.

8.3 Database

```
sudo mariadb
CREATE DATABASE glpi CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'glpi'@'localhost' IDENTIFIED BY '<strong password>';
GRANT ALL PRIVILEGES ON glpi.* TO 'glpi'@'localhost';
GRANT SELECT ON mysql.time_zone_name TO 'glpi'@'localhost';
FLUSH PRIVILEGES;
EXIT;
mariadb-tzinfo-to-sql /usr/share/zoneinfo | sudo mariadb mysql
```

8.4 GLPI files (secure layout)

Download the latest release archive from the GitHub releases page of **glpi-project/glpi** (replace X.Y.Z). Configuration and data are moved **outside** the web folder: that layout is also what the backup script saves.

```
cd /tmp
wget https://github.com/glpi-project/glpi/releases/download/X.Y.Z/glpi-X.Y.Z.tgz
sudo tar xzf glpi-X.Y.Z.tgz -C /var/www/html/
sudo mkdir -p /etc/glpi /var/lib/glpi /var/log/glpi
sudo mv /var/www/html/glpi/config/* /etc/glpi/
sudo mv /var/www/html/glpi/files/* /var/lib/glpi/
```

Create `/var/www/html/glpi/inc/downstream.php`:

```
<?php
define('GLPI_CONFIG_DIR', '/etc/glpi/');
if (file_exists(GLPI_CONFIG_DIR . '/local_define.php')) {
    require_once GLPI_CONFIG_DIR . '/local_define.php';
}
```

Create `/etc/glpi/local_define.php`:

```
<?php
define('GLPI_VAR_DIR', '/var/lib/glpi');
define('GLPI_LOG_DIR', '/var/log/glpi');
```

```
sudo chown -R root:root /var/www/html/glpi
sudo chown -R www-data:www-data /etc/glpi /var/lib/glpi /var/log/glpi
/var/www/html/glpi/marketplace
```

8.5 Apache virtual host

Create `/etc/apache2/sites-available/glpi.conf` — the document root must be **public**:

```
<VirtualHost *:80>
    ServerName lnx01.lab.local
    DocumentRoot /var/www/html/glpi/public
    <Directory /var/www/html/glpi/public>
        Require all granted
        RewriteEngine On
        RewriteCond %{HTTP:Authorization} ^(.+)$
        RewriteRule .* - [E=HTTP_AUTHORIZATION:%{HTTP:Authorization}]
        RewriteCond %{REQUEST_FILENAME} !-f
        RewriteRule ^(.*)$ index.php [QSA,L]
    </Directory>
</VirtualHost>
```

```
sudo a2enmod rewrite
sudo a2dissite 000-default
sudo a2ensite glpi
sudo sed -i 's/^session.cookie_httponly =.*/session.cookie_httponly = On/'
/etc/php/*/apache2/php.ini
sudo systemctl restart apache2
```

8.6 Web installer and hardening

1. From the host: **http://10.2.0.72** → language → licence → **Installer** → requirements check → SQL server `localhost`, user `glpi`, its password → database `glpi` → finish.
2. Log in **glpi** (default password glpi). Change its password; **disable** tech, normal, post-only (Administration → Utilisateurs).
3. `sudo rm /var/www/html/glpi/install/install.php`
4. Home page: **Désactiver les données de démonstration**.
5. **Configuration → Générale**: URL de l'application = `http://lnx01.lab.local`.

> **Watch out —** Right after a reboot, Apache and MariaDB take a moment: if GLPI does not answer, wait one minute before investigating.

8.7 Link GLPI to Active Directory (LDAP)

Prerequisite: account **svc-glpi** in COMPTES-ADMINISTRATION (normal user, password never expires) and `php -m | grep ldap` shows ldap. **Configuration → Authentification → Annuaires LDAP → Ajouter**:

Field	Value
Nom	AD lab.local
Serveur par défaut / Actif	Oui / Oui
Serveur / Port	srv-ad.lab.local / 389 — after chapter 14: **ldaps://srv-ad.lab.local / 636**
Filtre de connexion	`(&(objectClass=user)(objectCategory=person))`
BaseDN	`OU=UTILISATEURS,DC=lab,DC=local` (no spaces around commas)
DN du compte / mot de passe	`svc-glpi@lab.local` / its password
Champ de l'identifiant	samaccountname

> **Watch out —** The filter with the AD matching rule `userAccountControl:1.2.840.113556.1.4.803:=2` is rejected by this GLPI version (“Bad search filter”). The simple filter above works; disabled accounts still cannot log in because AD refuses them. No spaces anywhere in a filter or DN.

1. Tab **Tester**: the 5 steps run automatically on opening (F5 to rerun); step 5 “Chercher” must be green.
2. **Administration → Utilisateurs → Liaison annuaire LDAP → Importation de nouveaux utilisateurs** → Rechercher → select → Actions → Importer (or let users be created at first login).
3. Login page: choose **Source de connexion** = **AD lab.local** for AD users, **Base interne GLPI** for glpi.

Note — Port 389 sends the bind password in clear text. Chapter 14 adds the internal PKI and switches this link to **LDAPS (636)**, encrypted with the domain controller's certificate.

8.8 SSH by key, then firewall

1. On LNX01: `sudo adduser tech1` and `sudo usermod -aG sudo tech1`.
2. From the host (PowerShell): first login `ssh tech1@10.2.0.72` (answer yes), then `ssh-keygen -t ed25519 -C "tech1@admin-pc"` (answer **n** if a key already exists: reuse it).
3. Copy the public key (Windows has no ssh-copy-id):

```
type $env:USERPROFILE\.ssh\id_ed25519.pub | ssh tech1@10.2.0.72 "mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
```

4. `ssh tech1@10.2.0.72` → logs in without password. **Keep this session open.**
5. Create `/etc/ssh/sshd_config.d/10-novatech.conf` with `PermitRootLogin no` and `PubkeyAuthentication yes` → `sudo sshd -t` → `sudo systemctl restart ssh`.
6. Test in a **second** window. Only then add `PasswordAuthentication no` and `KbdInteractiveAuthentication no`, test the syntax, restart.

```
sudo sshd -T | grep -Ei 'permitrootlogin|passwordauthentication'   (both: no)
ssh -o PubkeyAuthentication=no tech1@10.2.0.72                       (Permission denied (publickey))
```

Firewall — allow SSH before enabling ufw:

```
sudo apt install -y ufw
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow from 10.2.0.96/28 to any port 22 proto tcp
sudo ufw allow 80/tcp
sudo ufw enable
sudo ufw status verbose
sudo ss -tlnp            (MariaDB 3306 on 127.0.0.1 only)
```

Check — From CLT01: GLPI opens; `Test-NetConnection 10.2.0.72 -Port 22` → False. From the host: SSH works with the key.

9. NAS01 — OpenMediaVault, RAID 1

9.1 VM and installation

Field	Value
VM	04 - NAS01 · Linux → Debian 64-bit · 2 vCPU · 1.5–2 GB · segment SRV20
Disks	16–20 GB system + **2 × 50 GB** data (for the mirror)
Installer	French; network manual: 10.2.0.73/27 · gw 10.2.0.65 · DNS 10.2.0.70 · nas01 / lab.local; root password; mirror France
Install disk	**the small system disk** — OMV takes the whole disk and cannot store data on it

1. On SRV-AD: A record **nas01 → 10.2.0.73** with PTR.
2. Browse **http://10.2.0.73**, log in `admin` / `openmediavault`, change the password immediately (user icon → Change Password).
3. **Système → Gestion des mises à jour**: install updates.
4. **Système → Date & heure**: Europe/Paris, NTP ✓ server `10.2.0.70`.

Watch out — OMV applies changes in two steps: **Save**, then the yellow **Apply** banner.

9.2 RAID 1, file system, share

1. **Stockage → Disques**: wipe **/dev/sdb** then **/dev/sdc** one at a time (eraser → Rapide). Never sda.
2. **Système → Plugins**: install **openmediavault-md**, F5.
3. **Stockage → Multiple Device → +**: niveau **Miroir**, devices sdb + sdc → Save → Apply; wait for state **clean**.
4. **Stockage → Systèmes de fichiers → + → EXT4** on **/dev/md0**, then mount it (warning threshold 85%).
5. **Stockage → Dossiers partagés → +**: Sauvegardes on md0, permissions *Administrateur r/w, Utilisateurs r/w, Autres: aucun*.
6. **Utilisateurs → +**: `SVC-Backup` (exact case), strong password (same as the domain SVC-Backup). Back in Dossiers partagés → Sauvegardes → **Permissions**: SVC-Backup lecture/écriture.
7. **Services → SMB/CIFS → Paramètres**: Activé, workgroup LAB (NetBIOS, no dot), home folders off. **Partages → +**: Sauvegardes, Public **Non**, read-only ✗, recycle bin optional. Apply.
8. From SRV-AD open `\\nas01\Sauvegardes` as `nas01\SVC-Backup`, create folders **SRV-AD, FS01, LNX01, OPNSENSE, SIEM01**.

Why — RAID 1 survives one disk failure; it is **not a backup**: a deleted file or ransomware is mirrored instantly. Linux user names are **case-sensitive**: SVC-Backup ≠ svc-backup.

9.3 SSH policy

Services → SSH: either disabled (smallest attack surface), or enabled with **root login off** and a dedicated admin user member of group **_ssh** (use `su -` once connected — sudo is not installed on OMV). Enable it temporarily when you need a shell (Wazuh agent install).

10. Backups and restore test

Machine	What	When	Tool	Target
SRV-AD	Full recovery (system state + C: + EFI)	21:00	Sauvegarde Windows Server	\\nas01\Sauvegardes\SRV-AD
FS01	D: only (data + share/ACL exports)	21:30	Sauvegarde Windows Server	\\nas01\Sauvegardes\FS01
LNX01	GLPI database + files + configs	22:00	bash script + cron	\\nas01\Sauvegardes\LNX01
OPNsense	config XML (encrypted)	after each change	GUI download	...\OPNSENSE
SIEM01	wazuh-install-files.tar	once after install	scp	...\SIEM01

10.1 Service account

A scheduled backup runs **under the account you enter**. A DC has no local accounts, so the account must exist in AD: **LAB\SVC-Backup** (member of **Opérateurs de sauvegarde**), with **exactly the same name and password** as the local SVC-Backup on the NAS. Samba then accepts it because name and password match.

> **Watch out —** Change one password → change the other, or every backup fails.

10.2 SRV-AD

1. PowerShell admin: `Install-WindowsFeature Windows-Server-Backup`.
2. `net use * /delete` (clear old connections to the NAS).
3. **wbadmin.msc → Sauvegarde locale → Planification de sauvegarde**: Personnalisée → **Récupération complète du système** → une fois par jour 21:00 → **Dossier réseau partagé** `\\nas01\Sauvegardes\SRV-AD` → identifiants `LAB\SVC-Backup`.
4. Accept the warning: a network share keeps **only the latest** copy.
5. **Sauvegarde unique → Options de sauvegarde planifiée → Sauvegarde** (≈ 22 GB, 10–20 min).

10.3 FS01

1. `lusrmgr.msc` → Groupes → **Opérateurs de sauvegarde** → add `LAB\SVC-Backup` (a member server has its own local group).
2. Install the feature (same command).
3. Export share definitions and ACLs into D:, so they are inside the backup (cmd admin):

```
mkdir D:\CONFIG
reg export "HKLM\SYSTEM\CurrentControlSet\Services\LanmanServer\Shares" D:\CONFIG\partages.reg /y
icacls D:\PARTAGES /save D:\CONFIG\acl-partages.txt /t /c
```

4. Schedule: Personnalisée → **DATA (D:) only** → 21:30 → `\\nas01\Sauvegardes\FS01` → LAB\SVC-Backup. Run a Sauvegarde unique.

> **Why —** System state of FS01 (≈ 17.5 GB) blocked VSS and would fill the 49 GB mirror. If FS01 dies: reinstall, rejoin the domain, restore D:, then `partages.reg` and `icacls D:\ /restore D:\CONFIG\acl-partages.txt`.

> **Watch out —** If a backup sits at 0 KB and the VM lags, VSS is stuck (often after a cancelled backup): `wbadmin stop job`, reboot, `vssadmin list writers` (all "Stable / Aucune erreur"), retry.

10.4 LNX01 — script and cron

```
sudo apt install -y cifs-utils
sudo nano /root/.smb-nas          # username=SVC-Backup / password=... / domain=nas01
sudo chmod 600 /root/.smb-nas
```

`/usr/local/sbin/backup-glpi.sh` (then `sudo chmod 700`):

```
#!/bin/bash
set -e
DATE=$(date +%F_%H%M)
MNT=/mnt/nas
DEST=$MNT/LNX01
mkdir -p $MNT
mount -t cifs //10.2.0.73/Sauvegardes $MNT -o credentials=/root/.smb-nas,vers=3.0
mkdir -p $DEST
mysqldump --single-transaction glpi | gzip > $DEST/glpi-db_$DATE.sql.gz
tar czf $DEST/glpi-files_$DATE.tar.gz --exclude=/var/lib/glpi/_cache \
  --exclude=/var/lib/glpi/_sessions --exclude=/var/lib/glpi/_tmp \
  /var/www/html/glpi /etc/glpi /var/lib/glpi /etc/apache2/sites-available /etc/ssh/sshd_config.d
find $DEST -name "glpi-*" -mtime +7 -delete
umount $MNT
echo "$(date) sauvegarde OK" >> /var/log/backup-glpi.log
```

```
sudo /usr/local/sbin/backup-glpi.sh && cat /var/log/backup-glpi.log
sudo crontab -e      # add:  0 22 * * * /usr/local/sbin/backup-glpi.sh
```

Why — The NAS is mounted only during the backup, so a compromised LNX01 cannot reach the backups the rest of the time. `/etc/glpi` holds **glpicrypt.key**: without it, stored passwords (LDAP bind...) are unreadable after a restore. `--single-transaction` gives a consistent dump without stopping GLPI.

10.5 Checks and restore test

Check	How
Job without error	wbadmin.msc → Messages: **Réussite**; `wbadmin get versions -backupTarget:\\nas01\Sauvegardes\SRV-AD`
Files present	\\nas01\Sauvegardes\SRV-AD\WindowsImageBackup\... · LNX01: glpi-db_... and glpi-files_...
Free space	OMV dashboard / Stockage → Systèmes de fichiers (alert at 85%)
LNX01 log	/var/log/backup-glpi.log ends with “sauvegarde OK”

1. On FS01 create `D:\PARTAGES\COMMUN\test-restauration.txt` with some text; run a Sauvegarde unique.
2. Delete the file **and empty the Recycle Bin**.
3. **wbadmin.msc → Récupérer...** → Une sauvegarde stockée à un autre emplacement → Dossier partagé distant → \\nas01\Sauvegardes\FS01 → today's version → **Fichiers et dossiers** → select the file → Emplacement d'origine, ✓ **Restaurer les autorisations ACL** → Récupérer.
4. The file is back with its text and its permissions (DL_FS_COMMUN_RW).

Why — A backup never restored proves nothing. Snapshot = fast rollback of one VM, stored on the same disk; backup = independent copy on another device.

11. SIEM01 — Wazuh

11.1 VM and Ubuntu installation

Field	Value
VM	03 - SIEM01 · Ubuntu 64-bit · **4 vCPU · 6–8 GB** · 60 GB · **VMnet1** (admin network)
Creation	“J'installerai le système d'exploitation plus tard” (avoids VMware Easy Install)
ISO	Ubuntu Server 26.04 (live-server)
Network	ens33 → Edit IPv4 → Manual: subnet 10.2.0.96/28 · address 10.2.0.102 · gateway 10.2.0.97 · name servers 10.2.0.70 · search lab.local
Storage	entire disk, LVM — then **ubuntu-lv → Edit → Size = max** (by default only half is used)
Profile / extras	server name siem01 · user nicolas · skip Ubuntu Pro · ✓ **Install OpenSSH server** · no snaps

Note — Wazuh's documentation lists Ubuntu up to 24.04; on 26.04 the installer prints a “not in the list of recommended systems” warning and installs fine (Wazuh 4.14.9). Check support before choosing an OS for a server product.

Watch out — If the mirror test fails with “Temporary failure resolving”, check that OPNsense and SRV-AD are **running** and that the IP settings were saved (card shows “static 10.2.0.102/28”).

1. On SRV-AD: A record **siem01 → 10.2.0.102** with PTR.
2. `sudo apt update && sudo apt full-upgrade -y` · `sudo timedatectl set-timezone Europe/Paris`.
3. Time from the DC — Ubuntu 26.04 uses **chrony** (not systemd-timesyncd):

```
echo "server 10.2.0.70 iburst prefer" | sudo tee /etc/chrony/sources.d/lab.sources
sudo systemctl restart chrony
chronyc sources          (SRV-AD becomes ^* or ^+ once the DC itself is synchronised)
```

11.2 Install Wazuh (all-in-one)

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

10–20 minutes. At the end it prints **User: admin / Password: …** — store it immediately. To read it again:

```
sudo tar -O -xvf wazuh-install-files.tar wazuh-install-files/wazuh-passwords.txt | grep -A1
"'admin'"
```

11.3 Post-install fixes

1. **Indexer memory**: the script sets 1 GB. Edit `/etc/wazuh-indexer/jvm.options`, change the **real** lines at the bottom (not the ## examples) to half the VM RAM: `-Xms4g / -Xmx4g` for 8 GB (3g for 6 GB), then `sudo systemctl restart wazuh-indexer`.
2. **Keep the install archive** (passwords, root CA and all private keys): copy `wazuh-install-files.tar` to your home, `scp` it to the host, put it in `\\nas01\Sauvegardes\SIEM01`, delete the host copy. Re-running `-a` deletes it.
3. **Freeze the version** (agents must never be newer than the manager):

```
sudo sed -i "s/^deb /#deb /" /etc/apt/sources.list.d/wazuh.list
sudo apt update
```

Dashboard: **https://10.2.0.102** → admin. Changing the admin password must be done with `/usr/share/wazuh-indexer/plugins/opensearch-security/tools/wazuh-passwords-tool.sh -u admin -p '…'` (never only in the GUI: Filebeat uses the same account).

11.4 Agents

☰ → **Agents management → Summary → Deploy new agent**: package, server `10.2.0.102`, agent name, group default → copy the generated command (it uses the server's version).

Agent	Package	Start	Network prerequisite
SRV-AD, FS01	Windows MSI	NET START Wazuh	SERVEURS → any (existing)
CLT01	Windows MSI	NET START Wazuh	rule UTILISATEURS → SIEM01:WAZUH_PORTS
LNX01, PROXY01	Linux DEB amd64	systemctl daemon-reload; enable --now wazuh-agent	PROXY01: rule DMZ → SIEM01
NAS01	Linux DEB amd64 (run as root, remove “sudo”)	idem	SSH or console temporarily

```
# example of generated Linux command
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.9-1_amd64.deb
&& \
  sudo WAZUH_MANAGER='10.2.0.102' WAZUH_AGENT_NAME='LNX01' dpkg -i
./wazuh-agent_4.14.9-1_amd64.deb
```

Check — Agents list: every agent **active**, same version as the server. “Disconnected” = VM off (or agent stopped — a real warning sign on a running machine).

11.5 OPNsense logs by syslog

In `/var/ossec/etc/ossec.conf` (backup first: `sudo cp ossec.conf ossec.conf.bak`) add a **second** remote block just after the agents' block:

```
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>10.2.0.97</allowed-ips>
  <local_ip>10.2.0.102</local_ip>
</remote>

sudo systemctl restart wazuh-manager
sudo ss -ulnp | grep 514                          (10.2.0.102:514 wazuh-remoted)
sudo tcpdump -ni any udp port 514 -c 5            (packets from 10.2.0.97)
```

Track A: OPNsense remote syslog (2A.8) and Log ✓ on block rules (2A.5). Track B: `logging host` on the router and switch (9B.2) — add 10.2.0.98 to allowed-ips.

Why — Rule 87701 (“pfSense firewall drop event”) has `<options>no_log</options>`: single blocks are counted but never stored. Rule **87702** (level 10, MITRE T1110) fires when the same source is blocked **18 times in 45 s**, then stays silent 240 s. OPNsense uses the pfSense log format, so Wazuh's built-in **pf** decoder works.

11.6 Detection tests

Test	Action	Expected in Threat Hunting → Events
Account lockout	on FS01: `runas /user:LAB\test.recette cmd` with a wrong password 5× → 6th gives error 1909	rule **60115** “User account locked out”, level 9, agent SRV-AD; filter `data.win.system.eventID: 4740`
Scan / repeated blocks	on CLT01: `ping -t -w 200 10.2.0.97` for 1 min	rule **87702**, level 10, agent siem01, srcip 10.2.0.10
DMZ server moving laterally	on PROXY01: 25 × `timeout 1 bash -c '</dev/tcp/10.2.0.70/445'`	rule 87702 with srcip 10.2.0.114
Noise filter	search `rule.level >= 7`	only significant alerts

Note — Unlock afterwards: `Unlock-ADAccount test.recette` (or Compte → Déverrouiller). The unlock box only appears in the account tab while the account is locked.

11.7 Diagnostic toolbox

Tool	Use
`<logall>yes</logall>` in ossec.conf	stores every received event in /var/ossec/logs/archives/archives.log — diagnostics only, switch back to no (fills the disk)
`sudo /var/ossec/bin/wazuh-logtest`	paste one log line (no leading spaces) → shows pre-decoding, decoder, rule, “Alert to be generated”
`grep 8770 /var/ossec/logs/alerts/alerts.json`	were alerts really created?
`/var/ossec/logs/ossec.log`	manager errors (remoted, allowed IPs…)

12. DMZ — publishing GLPI over HTTPS

The only server exposed to the “Internet” sits alone in the DMZ. If it is compromised, the attacker can only reach GLPI's port 80, the DNS and the SIEM agent port — everything else is blocked and logged.

```
Outside (host PC through VMware NAT, 192.168.10.x)
      |  HTTPS 443
[ OPNsense WAN 192.168.10.136 ]  Destination NAT 443 → 10.2.0.114
      |
DMZ 10.2.0.112/28 ── PROXY01 10.2.0.114 (Nginx, TLS)
      |  HTTP 80 only
SERVEURS ── LNX01 10.2.0.72 (GLPI)
```

Network parts are already done in your track. Track A: interface em4 (2A.2), aliases (2A.4), DMZ rules (2A.5), GUI on ADMIN only and WAN private networks allowed (2A.7), port forwarding below. Track B: sub-interface G0/0.40 (6B), ACL-DMZ (8B.2), static PAT 443 (7B) — skip 12.3.

12.1 PROXY01

Field	Value
VM	07 - PROXY01 · Debian 13 netinst · 1 vCPU · 1 GB · 10 GB · segment **DMZ40**
Network	manual: 10.2.0.114/28 · gw 10.2.0.113 · DNS 10.2.0.70 · proxy01 / lab.local
Root password	empty (sudo for the user)
Software	✗ desktop · ✓ SSH server · ✓ standard utilities

On SRV-AD: A record **proxy01 → 10.2.0.114** with PTR. Then check the isolation from PROXY01:

```
sudo apt update && sudo apt install -y curl                      (Internet 80/443: OK)
getent hosts lnx01.lab.local                                      (DNS: 10.2.0.72)
curl -I http://10.2.0.72                                          (HTTP/1.1 200 OK, Server:
Apache)
timeout 3 bash -c '</dev/tcp/10.2.0.70/445' && echo OPEN || echo BLOCKED   (BLOCKED)
timeout 3 bash -c '</dev/tcp/10.2.0.72/22'  && echo OPEN || echo BLOCKED   (BLOCKED)
```

12.2 Nginx with TLS

```
sudo apt install -y nginx
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
  -keyout /etc/ssl/private/glpi.key -out /etc/ssl/certs/glpi.crt \
  -subj "/CN=glpi.lab.local" \
  -addext "subjectAltName=DNS:glpi.lab.local,IP:192.168.10.136"
```

`/etc/nginx/sites-available/glpi`:

```
server {
    listen 80;
    server_name _;
    return 301 https://$host$request_uri;
}
server {
    listen 443 ssl;
    server_name _;
    ssl_certificate     /etc/ssl/certs/glpi.crt;
    ssl_certificate_key /etc/ssl/private/glpi.key;
    ssl_protocols       TLSv1.2 TLSv1.3;
    server_tokens off;
    location / {
        proxy_pass http://10.2.0.72;
        proxy_set_header Host              $host;
        proxy_set_header X-Real-IP         $remote_addr;
        proxy_set_header X-Forwarded-For   $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto https;
    }
}
```

```
sudo ln -s /etc/nginx/sites-available/glpi /etc/nginx/sites-enabled/
sudo rm /etc/nginx/sites-enabled/default
sudo nginx -t && sudo systemctl reload nginx
curl -kI https://localhost          (200 OK, "Server: nginx" without version, no Apache)
```

Why — `server_tokens off` and the proxy hide the real server version; X-Forwarded-* pass the visitor's IP and the HTTPS scheme to GLPI. The private key folder is readable by root only — “Permission non accordée” for a normal user is expected.

Note — The certificate contains the WAN IP **192.168.10.136**, obtained by DHCP from VMware. If it changes, regenerate the certificate — or give the WAN a static address (192.168.10.136/24, gateway 192.168.10.2).

Note — This self-signed certificate is the starting point. **Chapter 14.9** replaces it with a certificate issued by the internal CA (LAB-ROOT-CA): no more browser warning.

12.3 Port forwarding

Firewall → NAT → Destination NAT → +:

Field	Value
Description	Publication GLPI via PROXY01
Interface / Version / Protocol	WAN / IPv4 / TCP
Source	any
Destination / port	WAN address / HTTPS (443)
Redirect target IP / port	10.2.0.114 / HTTPS (443)
Firewall rule	the option that registers the associated rule (not “Manual”)
Log	✓ (optional)

Check — Firewall → Rules → WAN → Automatically generated rules: pass TCP → **10.2.0.114:443** with a chain-link icon (destination is the translated address: NAT happens before filtering). From the host: **https://192.168.10.136** → certificate warning → GLPI login → log in with an AD account. Certificate details show CN = glpi.lab.local.

Finally install the Wazuh agent on PROXY01 (chapter 11.4) — the exposed server is the one you most need to watch.

13. GLPI in use

13.1 Profiles and categories

1. **Administration → Utilisateurs → maxime.roux → Habilitations**: Entité racine, profil **Technician**, récursif Oui → Ajouter. Same for clara.garnier. (D = dynamic from LDAP, R = recursive.)
2. **Configuration → Intitulés → Catégories ITIL** → + : Utilisateur, Windows, Réseau, Serveur (optional: technicien responsable = maxime.roux).

13.2 Ticket workflow

Nouveau → En cours (Attribué) → Résolu → Clos. Résolu = the technician believes it is fixed; Clos = the requester approved (or automatic closure after a delay).

Ticket	Type / category / source	Requester → technician	Follow-up and solution
#1 Lecteur F: absent	Incident / Windows / Phone	emma.laurent → maxime.roux	gpresult /r OK, user in GG_COMPTABILITE → session opened before the GPO; logoff/logon remaps F:
#2 Accès refusé à \\FS01\Direction	Demande / Réseau / E-Mail	jean.boulet → maxime.roux	AGDLP check: only GG_COMMERCIAL → normal behaviour, access needs Direction's approval

1. Log in as maxime.roux (source AD lab.local), active profile **Technician**.
2. **Assistance → Créer un ticket**: remove Maxime from Demandeur, add the user; Attribué à = Roux Maxime (or the person icon “assign to me”) → Ajouter.
3. **Répondre ⌄ → Suivi**, optional **Tâche** (duration, statut Fait), then **Solution**.
4. Approve as requester (or set Statut **Clos**). Closed tickets are hidden by the default status filter: **Filtrer par Statut → Tous**.

> **Note —** Business users cannot log into GLPI outside the logon hours (AD enforces them on LDAP binds). Out of hours, the helpdesk logs the ticket on their behalf.

13.3 Knowledge base article

`http://10.2.0.72/front/knowbaseitem.form.php` (or Outils → Base de connaissances → Ajouter):

```
Sujet : Lecteur réseau absent après ouverture de session

Symptôme : un lecteur réseau (P:, F:, D:…) n'apparaît pas dans l'Explorateur.
Diagnostic :
1. Sur le poste, lancer gpresult /r et vérifier que GPO-UTIL-Lecteurs est appliquée.
2. Vérifier que l'utilisateur est membre du bon groupe global (ex. GG_COMPTABILITE pour F:).
3. Vérifier que le partage \\FS01\<partage> est accessible.
Solution :
- Fermer puis rouvrir la session.
- Si besoin : gpupdate /force puis nouvelle ouverture de session.
Ticket d'origine : #1
```

Then tab **Éléments liés → Ajouter → Ticket #1** to link it.

14. PKI — certificate authority, LDAPS and trusted certificates

Until now the lab had two weak points: GLPI sent the svc-glpi password to the domain controller **in clear text** (LDAP 389), and every web interface used a **self-signed** certificate (browser warnings). This chapter adds an internal **PKI** — the company's own certificate authority — to encrypt LDAP and issue trusted certificates, then puts the CA itself under SIEM monitoring.

14.1 Design

Choice	Value	Why
Hierarchy	**single tier**: one Enterprise Root CA	simple, integrated with AD (templates, auto-enrollment)
Server	**CA01**, dedicated (never on the domain controller)	a CA is a tier-0 asset; a CA on a DC is a classic audit finding
Name	**LAB-ROOT-CA**	appears in every certificate and CRL URL — permanent, no spaces
Key / hash / validity	RSA **4096** · SHA-256 · **10 years**	a root lives long; issued certificates are short (1 year)
Revocation	CRL (1 week) + delta CRL (1 day), published in AD (LDAP) and over **HTTP**	HTTP is needed by Linux, OPNsense and Wazuh
Certificates issued	domain controller (LDAPS) · glpi.lab.local · fw01.lab.local · siem01.lab.local	one per service, 1-year validity

Why — In production, mid-size and large organisations use **two tiers**: an **offline root** (powered off, often protected by an HSM) that only signs an online **issuing CA**. If the issuing CA is compromised, it is revoked and replaced without rebuilding trust everywhere. A single tier on a dedicated server is common and acceptable in small environments; an online responder (**OCSP**) can replace CRL downloads. Many companies now also use managed cloud PKI services instead of their own CA.

14.2 CA01 — VM and preparation

Field	Value
VM	09 - CA01 · Windows Server 2025 (Desktop Experience) · 2 vCPU · 2–3 GB · 60 GB · segment **SRV20** · folder C:\VMs\CA01 · library 01-Infrastructure
Name / IP	CA01 · 10.2.0.74/27 · gw 10.2.0.65 · DNS 10.2.0.70
Domain	member of lab.local, computer object in OU **SERVEURS**

1. VMware Tools → reboot · rename `CA01` → reboot · static IP · `nslookup lab.local`.
2. Join `lab.local` → reboot → log on as **LAB\Administrateur** (`whoami` must answer `lab\administrateur`).
3. On SRV-AD: move CA01 to OU SERVEURS; the A record ca01 → 10.2.0.74 is created automatically.

Watch out — Final name and domain membership before AD CS. Once the CA is installed, the server can no longer be renamed or moved to another domain. And configure AD CS **logged on as LAB\Administrateur** (member of Administrateurs de l'entreprise): with the local account, “AC d'entreprise” is greyed out and the wizard silently creates a **standalone** CA.

14.3 CAPolicy.inf

Created **before** the role, in `C:\Windows\CAPolicy.inf` (Notepad as administrator, type *Tous les fichiers* so it is not saved as .inf.txt):

```
[Version]
Signature="$Windows NT$"

[Certsrv_Server]
RenewalKeyLength=4096
RenewalValidityPeriod=Years
RenewalValidityPeriodUnits=10
```

```
CRLPeriod=Weeks
CRLPeriodUnits=1
CRLDeltaPeriod=Days
CRLDeltaPeriodUnits=1
LoadDefaultTemplates=0
```

LoadDefaultTemplates=0: the CA starts with no published template; only the ones needed are added (least privilege).

14.4 Install and configure AD CS

1. **Gérer → Ajouter des rôles et fonctionnalités → Services de certificats Active Directory** → role services **Autorité de certification** + **Inscription de l'autorité de certification via le Web** (installs IIS, which publishes the CRL over HTTP) → Installer.
2. Flag → **Configurer les services de certificats Active Directory**:

Screen	Value
Type d'installation	**AC d'entreprise**
Type d'AC	**AC racine**
Clé privée / chiffrement	new key · RSA#Microsoft Software Key Storage Provider · **4096** · **SHA256**
Nom de l'AC	**LAB-ROOT-CA** (suffix DC=lab,DC=local added automatically)
Validité	**10 ans** · database paths by default

Check — `certutil -cainfo type` → **0 — racine d'entreprise**. In certsrv.msc the **Modèles de certificats** folder exists (it is missing on a standalone CA).

14.5 Publication (CRL, AIA)

IIS blocks the "+" of the delta CRL file name (LAB-ROOT-CA+.crl). Allow it (cmd as administrator):

```
%windir%\system32\inetsrv\appcmd set config "Default Web Site/CertEnroll"
-section:system.webServer/security/requestFiltering -allowDoubleEscaping:true
```

1. certsrv.msc → LAB-ROOT-CA → Propriétés → **Extensions**: on the `http://…/CertEnroll/` lines, CDP has ✓ "Inclure dans les CRL… delta" and ✓ "Inclure dans l'extension CDP des certificats émis"; AIA has ✓ "Inclure dans l'extension AIA des certificats émis" (defaults).
2. Certificats révoqués → Toutes les tâches → **Publier** → Nouvelle liste de révocation.
3. `pkiview.msc`: every line **OK** (CA certificate, AIA #1/#2, CDP #1/#2, delta CRL #1/#2).

Check — On any domain member, after `gpupdate /force`: certlm.msc → Autorités de certification racines de confiance → **LAB-ROOT-CA** is present. An Enterprise CA publishes its certificate in AD; every domain machine trusts it without manual installation.

14.6 Certificate templates

Template	How	Used for
Authentification Kerberos	publish the existing template: Modèles de certificats → Nouveau → Modèle de certificat à délivrer	domain controller certificate → **LDAPS**
LAB-Serveur-Web	Modèles de certificats → Gérer → **Serveur Web** → Dupliquer le modèle, then publish it	HTTPS certificates (GLPI, OPNsense, Wazuh)

Tab (LAB-Serveur-Web)	Setting
Compatibilité — **first**	CA Windows Server 2016 · recipient Windows 10 / Server 2016 (enables the Key Storage Provider)
Général	name **LAB-Serveur-Web** · validity **1 year** · renewal 6 weeks
Traitement de la demande	✓ **Autoriser l'exportation de la clé privée**
Chiffrement / Nom du sujet / Sécurité	RSA 2048, SHA256 · **Fournir dans la demande** · defaults

Watch out — Do not click **Appliquer** before typing the name: the template is saved and its name is locked for good. Microsoft templates are never edited — always duplicated.

14.7 Auto-enrollment by GPO

1. Gestion de stratégie de groupe → lab.local → new GPO **GPO-DOMAINE-PKI-Inscription-Auto**, linked at the domain root.
2. Configuration ordinateur → Stratégies → Paramètres Windows → Paramètres de sécurité → **Stratégies de clé publique → Client des services de certificats – Inscription automatique**: Activé, ✓ renew expired / update pending / remove revoked, ✓ update certificates that use templates.
3. On SRV-AD: `gpupdate /force` then `certutil -pulse`.

Check — certsrv.msc → **Certificats délivrés**: request from **LAB\SRV-AD$**, template Authentification Kerberos, valid 1 year. The domain controller enrolled on its own and will renew on its own.

14.8 LDAPS for GLPI

The domain controller loads its new certificate and listens on **636**. Test on SRV-AD with `ldp` → Connexion → Se connecter → srv-ad.lab.local, port 636, ✓ SSL → expected *Host supports SSL, SSL cipher strength = 256 bits · Established connection*.

LNX01 is not a domain member: it must be taught to trust LAB-ROOT-CA (the CA serves its certificate in DER, Linux expects PEM):

```
cd /tmp
wget http://ca01.lab.local/CertEnroll/CA01.lab.local_LAB-ROOT-CA.crt
openssl x509 -inform DER -in CA01.lab.local_LAB-ROOT-CA.crt -out LAB-ROOT-CA.crt
sudo cp LAB-ROOT-CA.crt /usr/local/share/ca-certificates/
sudo update-ca-certificates                                  (1 added)
openssl s_client -connect srv-ad.lab.local:636 -brief </dev/null    (Verification: OK)
sudo systemctl restart apache2
```

GLPI → Configuration → Authentification → Annuaires LDAP → AD lab.local: Serveur `ldaps://srv-ad.lab.local`, Port **636**, “Utiliser TLS” unticked (that is STARTTLS on 389) → Sauvegarder → Tester.

Check — Tester: “Connexion à srv-ad.lab.local sur le port **636** réussie” and the 5 steps green; log in as an AD user (source AD lab.local). The bind password now travels inside TLS.

14.9 HTTPS certificate for GLPI (PROXY01)

Method used for every non-Windows device: the **private key is generated on the device and never leaves it**; only the request (CSR, public) goes to the CA.

```
sudo mkdir -p /etc/ssl/lab && cd /etc/ssl/lab
sudo openssl req -new -newkey rsa:2048 -nodes -keyout glpi.key -out glpi.csr -subj
"/CN=glpi.lab.local" -addext
"subjectAltName=DNS:glpi.lab.local,DNS:proxy01.lab.local,IP:192.168.10.136,IP:10.2.0.114"
sudo chmod 600 glpi.key
cat glpi.csr
```

On CA01: save the block as `C:\PKI\glpi.csr`, then (cmd as administrator):

```
certreq -submit -attrib "CertificateTemplate:LAB-Serveur-Web" C:\PKI\glpi.csr C:\PKI\glpi.cer
```

Open glpi.cer in Notepad, paste it on PROXY01 as `/etc/ssl/lab/glpi.crt` and check:

```
sudo openssl x509 -in glpi.crt -noout -subject -issuer -ext subjectAltName
sudo openssl x509 -in glpi.crt -noout -modulus | md5sum      (same value as the key)
sudo openssl rsa  -in glpi.key -noout -modulus | md5sum
```

In `/etc/nginx/sites-available/glpi`: `ssl_certificate /etc/ssl/lab/glpi.crt;` and `ssl_certificate_key /etc/ssl/lab/glpi.key;` → `sudo nginx -t && sudo systemctl reload nginx`. The old self-signed files stay in place for a rollback.

Note — PROXY01 cannot download the root certificate from CA01: the DMZ is not allowed to reach the lab — the isolation works. Copy the root certificate by hand (it is public) into `/usr/local/share/ca-certificates/LAB-ROOT-CA.crt`, run update-ca-certificates, then `curl -sSI https://glpi.lab.local --resolve glpi.lab.local:443:127.0.0.1` → HTTP 200 with no certificate error.

1. **Host PC** (not a domain member): browse `http://10.2.0.74/CertEnroll/CA01.lab.local_LAB-ROOT-CA.crt` → Installer un certificat → **Ordinateur local** → Autorités de certification racines de confiance.
2. Optional: add `192.168.10.136 glpi.lab.local` to `C:\Windows\System32\drivers\etc\hosts`.
3. Close every Edge window, then open https://192.168.10.136 (or https://glpi.lab.local): **padlock**, no warning.

14.10 OPNsense web interface

1. **System → Trust → Authorities** → + → Import an existing Certificate Authority → paste LAB-ROOT-CA's certificate, **no private key**.
2. **System → Trust → Certificates** → + → Method **Create a Certificate Signing Request** · Server Certificate · RSA-2048 · SHA256 · CN `fw01.lab.local` · DNS fw01.lab.local · IP 10.2.0.97.
3. Copy the CSR → sign it on CA01 with certreq (same command, fw01.csr / fw01.cer).
4. Edit the entry → Method **Import certificate (signed by CA)** → paste **fw01.cer** → Save (Issuer shows LAB-ROOT-CA).
5. **System → Settings → Administration** → SSL Certificate **fw01.lab.local** → Save.

Watch out — If the new entry shows an empty Issuer and a lifetime, it was created as **self-signed** (wrong method). And “Invalid X509 certificate” on import means the **.csr** was pasted instead of the **.cer**: the block must start with BEGIN CERTIFICATE, without “REQUEST”.

14.11 Wazuh dashboard

Wazuh keeps its own internal PKI between indexer, manager and dashboard — it is **not touched**. Only the certificate of the web interface (port 443) is replaced.

```
sudo mkdir -p /etc/ssl/lab && cd /etc/ssl/lab
sudo openssl req -new -newkey rsa:2048 -nodes -keyout siem01.key -out siem01.csr -subj
"/CN=siem01.lab.local" -addext "subjectAltName=DNS:siem01.lab.local,IP:10.2.0.102"
(sign siem01.csr on CA01, paste the result as /etc/ssl/lab/siem01.crt)
sudo cp /etc/ssl/lab/siem01.crt /etc/wazuh-dashboard/certs/lab-dashboard.pem
sudo cp /etc/ssl/lab/siem01.key /etc/wazuh-dashboard/certs/lab-dashboard-key.pem
sudo chown wazuh-dashboard:wazuh-dashboard /etc/wazuh-dashboard/certs/lab-dashboard.pem
/etc/wazuh-dashboard/certs/lab-dashboard-key.pem
sudo chmod 400 /etc/wazuh-dashboard/certs/lab-dashboard.pem
/etc/wazuh-dashboard/certs/lab-dashboard-key.pem
sudo cp /etc/wazuh-dashboard/opensearch_dashboards.yml
/etc/wazuh-dashboard/opensearch_dashboards.yml.bak
```

In `opensearch_dashboards.yml`, change only the file names of `server.ssl.key` (lab-dashboard-key.pem) and `server.ssl.certificate` (lab-dashboard.pem), then `sudo systemctl restart wazuh-dashboard` (30–60 s).

Watch out — Use full file names: in `sudo chown … lab-dashboard*.pem`, the * is expanded by **your** shell, which cannot list the protected certs folder — the command then fails with “No such file”. Rollback if the dashboard does not start: restore the .bak and restart.

14.12 Monitoring the CA with Wazuh

A CA is a tier-0 server: whoever controls it can issue a certificate in anyone's name. Two layers are needed, because CA auditing is **off by default**:

1. certsrv.msc → LAB-ROOT-CA → Propriétés → **Audit**: tick the 7 boxes → Appliquer.
2. Enable the matching audit subcategory (language-independent GUID):

```
auditpol /set /subcategory:"{0CCE9221-69AE-11D9-BED3-505054503030}" /success:enable
/failure:enable
```

3. Install the Wazuh agent on CA01 (Deploy new agent → Windows MSI → name CA01) → `NET START Wazuh`.
4. The events reach Wazuh but no built-in rule raises them. Add custom rules on SIEM01 at the end of `/var/ossec/etc/rules/local_rules.xml` (back it up first):

```
<group name="windows,adcs,">
  <rule id="100200" level="5">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4880$|^4881$</field>
    <description>AD CS: Certificate Services started or stopped</description>
  </rule>
  <rule id="100201" level="5">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4887$</field>
    <description>AD CS: certificate issued by the CA</description>
  </rule>
  <rule id="100202" level="10">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4882$|^4885$|^4890$|^4899$|^4900$</field>
    <description>AD CS: CA security, configuration or template changed</description>
    <mitre><id>T1649</id></mitre>
  </rule>
  <rule id="100203" level="10">
    <if_sid>60103</if_sid>
    <field name="win.system.eventID">^4876$|^4877$</field>
    <description>AD CS: CA database and keys backed up or restored</description>
  </rule>
</group>
```

```
sudo /var/ossec/bin/wazuh-analysisd -t        (no output = syntax OK)
sudo systemctl restart wazuh-manager
```

Check — On CA01: `net stop certsvc && net start certsvc` → Threat Hunting, `rule.groups: adcs` → two alerts **100200**, agent CA01. IDs 100000–120000 are reserved for custom rules; T1649 = MITRE “Steal or Forge Authentication Certificates”.

14.13 Back up the CA

```
mkdir C:\CABackup
certutil -backup C:\CABackup                        (asks for a password protecting the .p12)
reg export HKLM\SYSTEM\CurrentControlSet\Services\CertSvc\Configuration
C:\CABackup\CertSvc-config.reg
copy C:\Windows\CAPolicy.inf C:\CABackup\
net use \\nas01\Sauvegardes /user:nas01\SVC-Backup *
mkdir \\nas01\Sauvegardes\CA01
robocopy C:\CABackup \\nas01\Sauvegardes\CA01 /E /R:2 /W:5
net use * /delete
```

Why — Without a backup, losing CA01 means rebuilding the whole PKI and redeploying trust everywhere. `/R:2 /W:5` stops robocopy from retrying a million times every 30 s when the share is unreachable. Store the .p12 password in the password manager (Lab › CA01 › CA backup).

14.14 Checks

Check	Expected
pkiview.msc	all OK
Certificats délivrés	SRV-AD$ (Kerberos), glpi, fw01, siem01 (LAB-Serveur-Web)
ldp on 636 / GLPI Tester	SSL 256 bits / port 636, 5 steps green
https://192.168.10.136 · https://10.2.0.97 · https://10.2.0.102	padlock, chain LAB-ROOT-CA → service
Wazuh rule.groups: adcs	alerts from agent CA01
\\nas01\Sauvegardes\CA01	LAB-ROOT-CA.p12, DataBase, CertSvc-config.reg, CAPolicy.inf

Part 5 — Operations

Acceptance tests, day-to-day procedures, known problems and improvement ideas.

15. Acceptance tests (recette)

Run this list after a rebuild. Fill the “Result” column and keep a screenshot of each proof. Track B: run the network tests of 11B first; tests 3–5 then refer to the router ACLs instead of OPNsense.

#	Area	Test	Expected	Result
1	Network	CLT01 gets an address	10.2.0.10–.60, DHCP server 10.2.0.70 (relay)	
2	Network	CLT01 pings servers by name	srv-ad, fs01, lnx01 answer	
3	Security	CLT01 → 10.2.0.97 / 10.2.0.1 GUI	blocked (logged in Live View)	
4	Security	CLT01 → SIEM01 :1514 / :443	True / False	
5	Admin	Host PC opens OPNsense, Wazuh, OMV	all three load	
6	AD	dcdiag /q on SRV-AD	no output	
7	Time	w32tm /query /status on SRV-AD	source pool.ntp.org, stratum 2–4	
8	DNS	nslookup 10.2.0.72 10.2.0.70	lnx01.lab.local	
9	Policy	net accounts	12 / 5 / 5 / 15 / 15	
10	Policy	reset a password to 9 characters	refused	
11	Policy	5 wrong passwords on test.recette	6th attempt: error 1909, account locked	
12	GPO	logon as Compta / Direction / Info	drives P F · P D F V T · P I	
13	GPO	wallpaper	applied	
14	GPO	cmd as business user / Informatique	blocked / allowed (gpresult /r)	
15	Hours	business logon outside Mon–Fri 08–17	refused (Windows and GLPI)	
16	AGDLP	emma.laurent → Direction	accès refusé	
17	AGDLP	alice.martin → Comptabilite, create a file	read OK, write refused	
18	GLPI	login AD account (maxime.roux)	OK, profiles Technician + Self-Service	
19	SSH	ssh tech1@10.2.0.72 / without key / from CLT01	OK / Permission denied (publickey) / blocked	
20	Backup	last job of SRV-AD, FS01; LNX01 log	Réussite · “sauvegarde OK”	
21	Restore	deleted test file restored	file back with ACLs	
22	SIEM	agents list	all active, same version	
23	SIEM	lockout test	rule 60115 on SRV-AD	
24	SIEM	fast ping from CLT01 to 10.2.0.97	rule 87702, srcip 10.2.0.10	
25	DMZ	PROXY01 → 10.2.0.70:445 and 10.2.0.72:22	BLOCKED	
26	DMZ	https://192.168.10.136 from the host	GLPI login page through Nginx	
27	DMZ	curl -kI https://localhost on PROXY01	Server: nginx (no version)	
28	PKI	pkiview.msc on CA01	every line OK (CA certificate, AIA, CDP, delta CRL)	
29	PKI	certlm.msc on SRV-AD → Personnel	certificate from LAB-ROOT-CA, template Kerberos Authentication	
30	PKI	GLPI LDAP directory → Tester	port 636, 5 steps green	
31	PKI	https://192.168.10.136 · https://10.2.0.97 · https://10.2.0.102 from the host	padlock, no warning	
32	PKI	net stop certsvc && net start certsvc on CA01	Wazuh rule 100200 (agent CA01)	

16. Operating procedures

16.1 New employee (arrival)

1. GLPI ticket (Demande, category Utilisateur).
2. dsa.msc → service OU → Nouveau → Utilisateur (prenom.nom, temporary password, ✓ “doit changer le mot de passe”).
3. Add to the service global group (GG_...) — **never** give folder permissions directly.
4. Business service: set logon hours Mon–Fri 08–17.
5. Test on CLT01: logon, drives, wallpaper; GLPI login (AD source).
6. Close the ticket.

16.2 Change of department (access change)

1. Ticket with the manager's approval.
2. Remove from old GG, add to new GG.
3. Move the account to the new service OU (GPO targeting and CMD policy follow the OU).
4. User logs off/on (Kerberos token refreshed); check access with **Accès effectif** on FS01.

16.3 Departure

1. Ticket. Disable the account (do not delete yet).
2. Remove from GG groups, reset the password.
3. Move to a disabled-accounts OU; archive personal data; delete after the retention period.
4. GLPI: the account can no longer authenticate (AD refuses it); deactivate it in GLPI as well.

16.4 Restore

Need	Procedure
A deleted file on FS01	wbadmin.msc → Récupérer → Fichiers et dossiers (chapter 10.5)
Share definitions / ACLs	import D:\CONFIG\partages.reg · icacls D:\ /restore D:\CONFIG\acl-partages.txt
GLPI	reinstall packages, extract glpi-files_*.tar.gz at /, `gunzip -c glpi-db_*.sql.gz | sudo mariadb glpi`, restart Apache (glpicrypt.key comes with /etc/glpi)
OPNsense	fresh install → System → Configuration → Backups → Restore → XML + password
Wazuh	certificates and passwords from wazuh-install-files.tar
A whole VM after a bad change	VMware snapshot → Revenir à l'instantané
The DC	Windows Server Backup bare-metal / system state recovery (one-DC lab: a snapshot rollback is also acceptable)
The CA	new server with the **same name**, AD CS role, then `certutil -restore` from \\nas01\Sauvegardes\CA01 (key .p12 + database) and import CertSvc-config.reg (chapter 14.13)

16.5 GLPI ticket

Create (type, category, source, requester, urgency) → assign → follow-ups / tasks with duration → solution → approval or closure → knowledge-base article if the problem can come back.

16.6 Certificates

Task	When / how
CRL validity	the full CRL is valid **1 week**: start CA01 at least once a week (it republishes on its own), or run `certutil -crl` after a long shutdown
Domain controller certificate	renewed automatically by the auto-enrollment GPO

Task	When / how
Web certificates (GLPI, OPNsense, Wazuh)	valid 1 year: new CSR on the device + certreq on CA01 about 6 weeks before expiry
Revoke a certificate	certsrv.msc → Certificats délivrés → right-click → Révoquer, then publish a new CRL
CA backup	after any change on the CA: certutil -backup + copy to the NAS (chapter 14.13)

17. Known problems and fixes

Real issues met while building this lab — useful as diagnosis examples.

Symptom	Cause	Fix
Host PC cannot reach 10.2.0.70; reply from 212.x.x.x “destination network unreachable”	no route to the lab: traffic leaves through the home box	route -p add 10.2.0.0 mask 255.255.255.128 10.2.0.97
Nothing answers in the lab after rebooting the laptop	VMs off (OPNsense / SRV-AD not started)	start order OPNsense → SRV-AD → others; wait
OPNsense WAN has no IP after install	WAN assigned to the host-only card	reassign interfaces using the MAC table
Disk added to a VM invisible (Proxmox, orange)	pending hot-plug change	full shutdown/start from the hypervisor
D: unavailable on FS01	CD-ROM holds D:	move the CD-ROM to E:
“Interface inconnue”, NETLOGON 5774, GroupPolicy 1054 after changing the DC's IP	AD/DNS still bound to the old IP	ipconfig /registerdns, restart netlogon, delete old A records, reboot the DC
DHCP event 1041; firewall profile “Privé” on the DC	the DC took its IP by DHCP	static IP before promotion
FS01 does not answer ping/SMB (Server 2025)	File and Printer Sharing rules disabled	Enable-NetFirewallRule -Group "@FirewallAPI.dll,-28502"
Local account refused after domain join	logon screen targets the domain	use .\Administrateur
Administrateur gets “accès refusé” on shares	by design (DL groups only)	use \\FS01\D$ or work locally
gpupdate hangs; wallpaper not applied	Group Policy client stuck	check DC/SYSVOL/time, then restart the client
Alias rejects 1514 / 67	alias type left on Host(s)	choose Port(s) first
GLPI login refused for a business user in the evening	AD logon hours apply to LDAP	test in hours or use Informatique accounts
GLPI LDAP “Bad search filter (-7)”	matching-rule filter rejected / spaces in DN	(&(objectClass=user)(objectCategory=person)), DN without spaces
GLPI does not load right after boot	Apache/MariaDB still starting	wait one minute
lnx01.lab.local unknown from the host PC	host uses the home box as DNS	use IPs from the host
Scheduled backup on the DC refuses nas01\SVC-Backup	a DC has no local accounts	domain account LAB\SVC-Backup with the same name and password as on the NAS
Share login refused	Linux user names are case-sensitive	type SVC-Backup exactly
FS01 backup stuck at 0 KB, VM laggy	system state / VSS stuck	wbadmin stop job, reboot, back up D: only
Backup files dated wrong on the NAS	NAS clock 12 h off	OMV NTP → 10.2.0.70, Europe/Paris
chrony marks SRV-AD “^?”	DC on its own CMOS clock	w32tm manualpeerlist pool.ntp.org /reliable:yes
systemd-timesyncd not found (Ubuntu 26.04)	chrony is the time service	/etc/chrony/sources.d/lab.sources
Ubuntu mirror test fails	router/DC not started or IP not saved	start OPNsense + SRV-AD, check static config
Wazuh indexer slow / crashing	heap left at 1 GB	jvm.options -Xms/-Xmx = half the RAM
No OPNsense alerts in Wazuh	87701 is no_log; block rules not logging	Log ✓ on block rules; look for 87702
OPNsense logs flooded by devd cd0	empty virtual CD drive	remove the CD drive from the VM
wazuh-logtest: “No decoder matched”	pasted line starts with spaces	paste the raw line starting with the date
Agent “disconnected”	VM off, or agent stopped	start the VM / service; investigate if running
PROXY01 → SIEM01 blocked	SIEM01 off, or Apply not clicked	start SIEM01, apply the rules
RAM at 97 % with all VMs	VMs reserve memory; iGPU reserved 4 GB	run by scenario; UMA buffer 1G in BIOS
CA created as **standalone**, no “Modèles de certificats” folder	AD CS configured while logged on with the local administrator	Uninstall-AdcsCertificationAuthority, log on as LAB\Administrateur, reconfigure as Enterprise CA
Delta CRL (LAB-ROOT-CA+.crl) not downloadable over HTTP	IIS blocks “+” in URLs	appcmd ... requestFiltering -allowDoubleEscaping:true on CertEnroll
Template name cannot be changed	Appliquer clicked before naming: the name is locked once saved	delete the copy, duplicate again, set Compatibility then name before OK
PROXY01 cannot download the root certificate from CA01	by design: the DMZ cannot reach the lab	copy the public root certificate by hand
OPNsense: “Invalid X509 certificate”	the request (.csr) pasted instead of the signed certificate (.cer)	paste the block BEGIN CERTIFICATE (no “REQUEST”)

Symptom	Cause	Fix
chown/chmod: “No such file or directory” with lab-dashboard*.pem	the * is expanded by the user's shell, which cannot list the protected folder	use full file names
CA events (4880, 4887…) absent from Wazuh	CA auditing off by default, then no built-in Wazuh rule	CA Audit tab + auditpol, custom rules 100200–100203

18. Snapshot and backup policy

	Snapshot (VMware)	Backup
Where	same SSD as the VM	NAS01 (RAID 1) — another device
Purpose	rollback of one VM in seconds	rebuild after loss or deletion
If the SSD dies	lost	still available
Rule	before every risky change; 1–2 per VM; delete old ones	nightly; restore tested

- Take snapshots with VMs **shut down** (smaller, faster). Names: `LAB-COMPLET-YYYY-MM-DD`, `AVANT-<change>`.
- Copy the VM folders to an external drive from time to time — the only copy that survives the laptop.
- Re-export the OPNsense configuration after every change.
- Moving the lab to another PC: copy the shut-down VM folders, open each .vmx, answer **“Je l'ai déplacée”** (keeps MACs), recreate VMnet1 settings, the host IP and the route; the encrypted CLT01 needs its VM password.

19. Improvement ideas

Idea	Benefit
Two-tier PKI (offline root + issuing CA), OCSP responder	production-grade PKI design (chapter 14.1)
Second DC (Server Core)	AD/DNS redundancy, replication practice
Off-site / external backup copy (3-2-1 rule)	survives loss of the laptop
Versioned backups (Veeam Agent, restic)	Windows Server Backup keeps only the last copy on a share
Public certificate (Let's Encrypt) on PROXY01	only if GLPI is ever published on a real domain name
Static WAN address on OPNsense	certificate SAN never breaks
LAPS, AppLocker / PowerShell restrictions	local admin passwords, block PowerShell for business users
Wazuh e-mail alerts, active response	notifications, automatic blocking
Weekly backup of /var/ossec/etc to the NAS	SIEM configuration and custom rules safe
CA01 in the Windows Server Backup schedule	automatic CA backups (system state includes AD CS)
Disable SSH root and passwords everywhere; keys only	consistent hardening
Desktop host with 64 GB RAM, or a mini PC with Proxmox	run the whole lab permanently; realistic VM backups

Appendices

Appendix A — Complete configurations

Ready to paste on factory-reset devices (replace the <secrets>). Same content as chapters 4B–9B.

A.1 SW-LAB

```
enable
configure terminal
hostname SW-LAB
no ip domain-lookup
ip domain-name lab.local
enable secret <enable-secret>
service password-encryption
username admin privilege 15 secret <admin-secret>
banner motd # Acces reserve aux administrateurs du lab #
clock timezone CET 1 0
clock summer-time CEST recurring last Sun Mar 2:00 last Sun Oct 3:00
service timestamps log datetime msec localtime show-timezone
spanning-tree mode rapid-pvst
vlan 10
 name UTILISATEURS
vlan 20
 name SERVEURS
vlan 30
 name ADMINISTRATION
vlan 40
 name DMZ
vlan 999
 name PARKING
interface range fa0/1 - 4
 description PC-ADMIN
 switchport mode access
 switchport access vlan 30
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
interface range fa0/5 - 8
 description PC-UTILISATEUR
 switchport mode access
 switchport access vlan 10
 switchport nonegotiate
 spanning-tree portfast
 spanning-tree bpduguard enable
 switchport port-security
 switchport port-security maximum 2
 switchport port-security violation restrict
 switchport port-security mac-address sticky
interface range fa0/9 - 24
 description NON-UTILISE
 switchport mode access
 switchport access vlan 999
 shutdown
interface range gi0/1 - 2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,40
 switchport nonegotiate
interface gi0/1
 description TRUNK-ROUTEUR
interface gi0/2
 description TRUNK-PROXMOX
interface vlan 1
 shutdown
interface vlan 30
 description MANAGEMENT
 ip address 10.2.0.98 255.255.255.240
 no shutdown
ip default-gateway 10.2.0.97
ntp server 10.2.0.70
```

```
logging host 10.2.0.102
logging trap informational
logging source-interface vlan 30
access-list 30 permit 10.2.0.96 0.0.0.15
line console 0
 login local
 logging synchronous
 exec-timeout 10 0
line vty 0 15
 transport input ssh
 login local
 access-class 30 in
 exec-timeout 10 0
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

A.2 R-LAB

```
enable
configure terminal
hostname R-LAB
no ip domain-lookup
ip domain-name lab.local
enable secret <enable-secret>
service password-encryption
username admin privilege 15 secret <admin-secret>
banner motd # Acces reserve aux administrateurs du lab #
no ip http server
no ip http secure-server
clock timezone CET 1 0
clock summer-time CEST recurring last Sun Mar 2:00 last Sun Oct 3:00
service timestamps log datetime msec localtime show-timezone
!
interface g0/0
 description TRUNK-SWITCH
 no ip address
 no shutdown
interface g0/0.10
 description VLAN10-UTILISATEURS
 encapsulation dot1Q 10
 ip address 10.2.0.1 255.255.255.192
 ip helper-address 10.2.0.70
 ip nat inside
interface g0/0.20
 description VLAN20-SERVEURS
 encapsulation dot1Q 20
 ip address 10.2.0.65 255.255.255.224
 ip nat inside
interface g0/0.30
 description VLAN30-ADMINISTRATION
 encapsulation dot1Q 30
 ip address 10.2.0.97 255.255.255.240
 ip nat inside
interface g0/0.40
 description VLAN40-DMZ
 encapsulation dot1Q 40
 ip address 10.2.0.113 255.255.255.240
 ip nat inside
interface g0/1
 description WAN-BOX
 ip address dhcp
 ip nat outside
 no shutdown
!
ip access-list standard NAT-LAB
 permit 10.2.0.0 0.0.0.255
ip nat inside source list NAT-LAB interface g0/1 overload
```

```
ip nat inside source static tcp 10.2.0.114 443 interface g0/1 443
!
ip access-list extended ACL-USERS
 permit udp any any eq bootps
 permit tcp 10.2.0.0 0.0.0.63 any established
 permit icmp 10.2.0.0 0.0.0.63 any echo-reply
 permit tcp 10.2.0.0 0.0.0.63 host 10.2.0.102 range 1514 1515
 deny   ip 10.2.0.0 0.0.0.63 10.2.0.96 0.0.0.15 log
 deny   tcp 10.2.0.0 0.0.0.63 host 10.2.0.1 eq 22 log
 permit ip 10.2.0.0 0.0.0.63 any
ip access-list extended ACL-DMZ
 permit tcp any any established
 permit udp host 10.2.0.114 host 10.2.0.70 eq domain
 permit tcp host 10.2.0.114 host 10.2.0.70 eq domain
 permit tcp host 10.2.0.114 host 10.2.0.72 eq www
 permit tcp host 10.2.0.114 host 10.2.0.102 range 1514 1515
 deny   ip any 10.2.0.0 0.0.0.255 log
 permit tcp host 10.2.0.114 any eq www
 permit tcp host 10.2.0.114 any eq 443
 permit udp host 10.2.0.114 any eq ntp
 deny   ip any any log
interface g0/0.10
 ip access-group ACL-USERS in
interface g0/0.40
 ip access-group ACL-DMZ in
!
ntp server 10.2.0.70
logging host 10.2.0.102
logging trap informational
logging source-interface g0/0.30
access-list 30 permit 10.2.0.96 0.0.0.15
line console 0
 login local
 logging synchronous
 exec-timeout 10 0
line vty 0 4
 transport input ssh
 login local
 access-class 30 in
 exec-timeout 10 0
crypto key generate rsa modulus 2048
ip ssh version 2
end
copy running-config startup-config
```

Appendix B — Variant: layer-3 switch (CCNA)

With a Catalyst 3560/3650, inter-VLAN routing moves to the switch (SVIs); the router keeps only NAT and the WAN. The router–switch link becomes a routed point-to-point link.

```
! ---- L3 switch
ip routing
interface vlan 10
 ip address 10.2.0.1 255.255.255.192
 ip helper-address 10.2.0.70
 ip access-group ACL-USERS in
interface vlan 20
 ip address 10.2.0.65 255.255.255.224
interface vlan 30
 ip address 10.2.0.97 255.255.255.240
interface vlan 40
 ip address 10.2.0.113 255.255.255.240
 ip access-group ACL-DMZ in
interface gi0/1
 description LIEN-ROUTEUR
 no switchport
 ip address 10.255.255.2 255.255.255.252
ip route 0.0.0.0 0.0.0.0 10.255.255.1

! ---- router
interface g0/0
 ip address 10.255.255.1 255.255.255.252
 ip nat inside
ip route 10.2.0.0 255.255.255.0 10.255.255.2
```

Note — Same ACLs as chapter 8B, applied on the SVIs. NAT-LAB stays on the router (it still matches 10.2.0.0/24). Compare both designs: router-on-a-stick is cheaper; SVIs route at wire speed and scale better.

Appendix C — Variant: OPNsense on hardware

You can also keep the exact VMware firewall configuration: run OPNsense as a Proxmox VM (or on a small PC with two NICs) and give it **VLAN interfaces** instead of separate cards:

1. OPNsense WAN on its own NIC (or on a dedicated VLAN to the ISP box).
2. **Interfaces → Devices → VLAN**: create VLANs 10, 20, 30, 40 on the LAN NIC (parent = the trunk side).
3. **Interfaces → Assignments**: assign each VLAN device, then use the same IPs, aliases, rules, relay, NAT and syslog as in chapter 2A.
4. The Cisco switch keeps the trunk/access configuration of chapter 5B; the router is no longer needed.

> **Why —** This variant is what most small companies run: a stateful firewall with a GUI, a managed switch for the VLANs, and the hypervisor on a trunk.

Appendix D — Cisco show commands

Goal	Command
Interfaces summary	show ip interface brief · show interfaces status
VLANs and trunks	show vlan brief · show interfaces trunk · show interfaces gi0/1 switchport
Spanning tree	show spanning-tree vlan 10 · show spanning-tree summary
Neighbours / cabling	show cdp neighbors detail · show lldp neighbors
MAC table	show mac address-table · show mac address-table interface fa0/5
Port security / errors	show port-security · show interfaces status err-disabled
Routing	show ip route · show ip protocols
DHCP relay	show run interface g0/0.10 · debug ip dhcp server packet (careful)
NAT	show ip nat translations · show ip nat statistics · clear ip nat translation *
ACLs	show access-lists · show ip interface g0/0.10
Management	show ip ssh · show users · show run | section line
Time and logs	show clock detail · show ntp associations · show logging
Configuration	show running-config · show startup-config · copy run start · show archive

Appendix E — Command cheat sheet (Windows / Linux)

Goal	Command
IP configuration (Windows)	ipconfig /all · ipconfig /renew · ipconfig /flushdns · ipconfig /registerdns
MAC addresses	Windows: getmac /v · Linux: ip -br link
Route table (host)	route print 10.2.0.0
Test a port	Windows: Test-NetConnection <ip> -Port <n> · Linux: timeout 3 bash -c '</dev/tcp/<ip>/<n>'
Domain controller health	dcdiag /q · nltest /dsgetdc:lab.local · repadmin /replsummary
Group Policy	gpupdate /force · gpresult /r · gpresult /h rapport.html
Account policy / lockout	net accounts · Unlock-ADAccount <user> · Get-ADUser <u> -Properties LockedOut,badPwdCount
Time (Windows / Linux)	w32tm /query /status · w32tm /resync · chronyc sources · timedatectl
Backups (Windows)	wbadmin get versions -backupTarget:<share> · wbadmin stop job · vssadmin list writers
Services (Linux)	systemctl status <svc> --no-pager · journalctl -u <svc> -n 50
Listening ports (Linux)	ss -tlnp · ss -ulnp
Firewall (Linux)	ufw status verbose
SSH effective config	sshd -T | grep -Ei 'permitroot|passwordauth'
Wazuh	systemctl status wazuh-manager · /var/ossec/bin/wazuh-logtest · tail /var/ossec/logs/ossec.log
Packet capture	tcpdump -ni any udp port 514 -c 5 [-A]
Nginx	nginx -t · systemctl reload nginx · curl -kI https://localhost
PKI: CA state	certutil -cainfo type · pkiview.msc · certsrv.msc
PKI: force enrollment / CRL	certutil -pulse (client) · certutil -crl (CA)
PKI: sign a CSR	certreq -submit -attrib "CertificateTemplate:LAB-Serveur-Web" req.csr cert.cer
PKI: back up the CA	certutil -backup C:\CABackup
TLS check (Linux)	openssl s_client -connect host:636 -brief · openssl x509 -in cert.crt -noout -subject -issuer
Trust a root (Debian)	copy .crt to /usr/local/share/ca-certificates/ · update-ca-certificates